

BFS Traversal

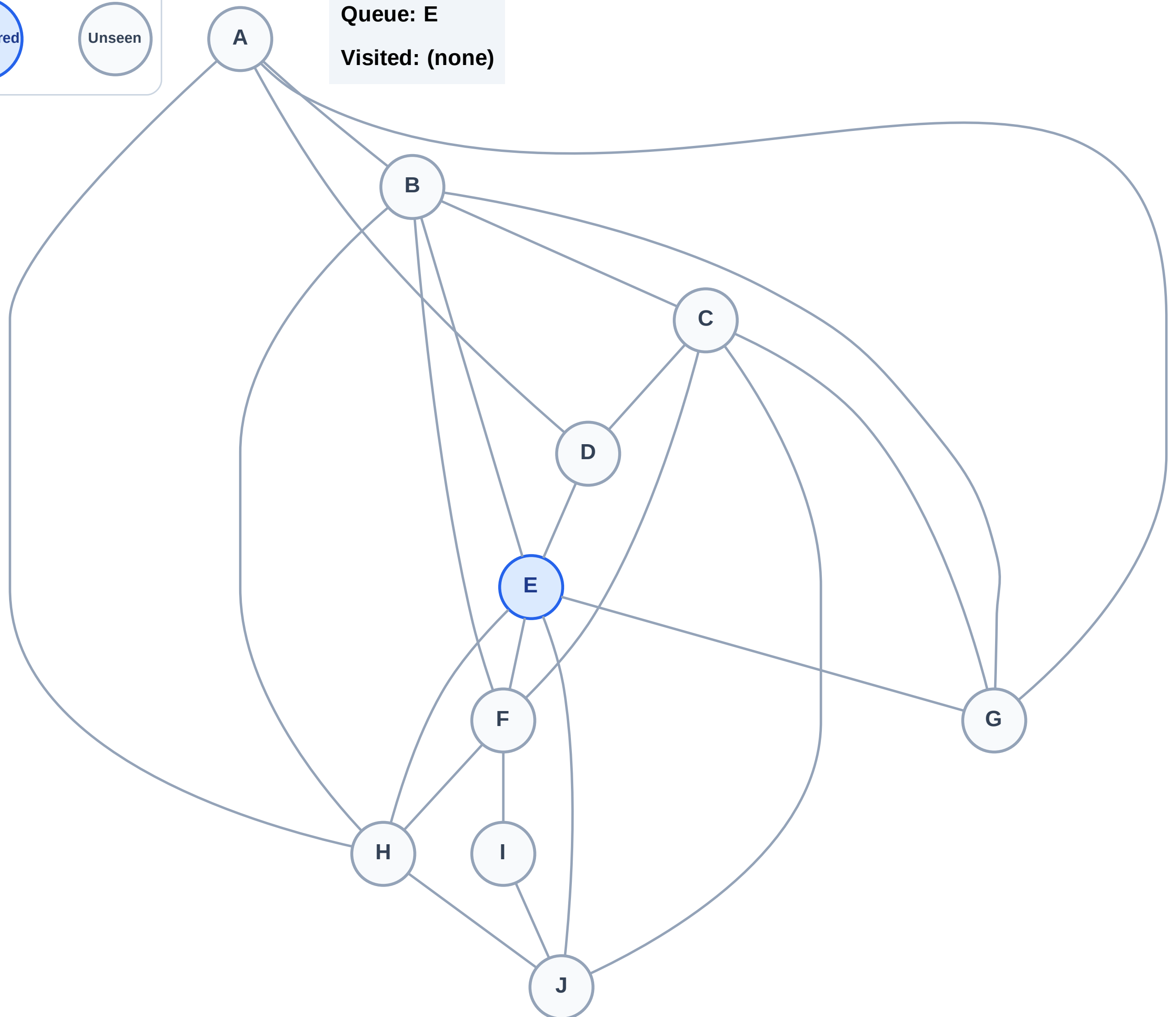
Step 1: Start at E

Legend



Queue: E

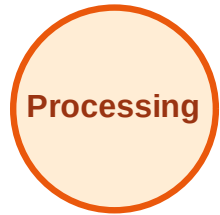
Visited: (none)



BFS Traversal

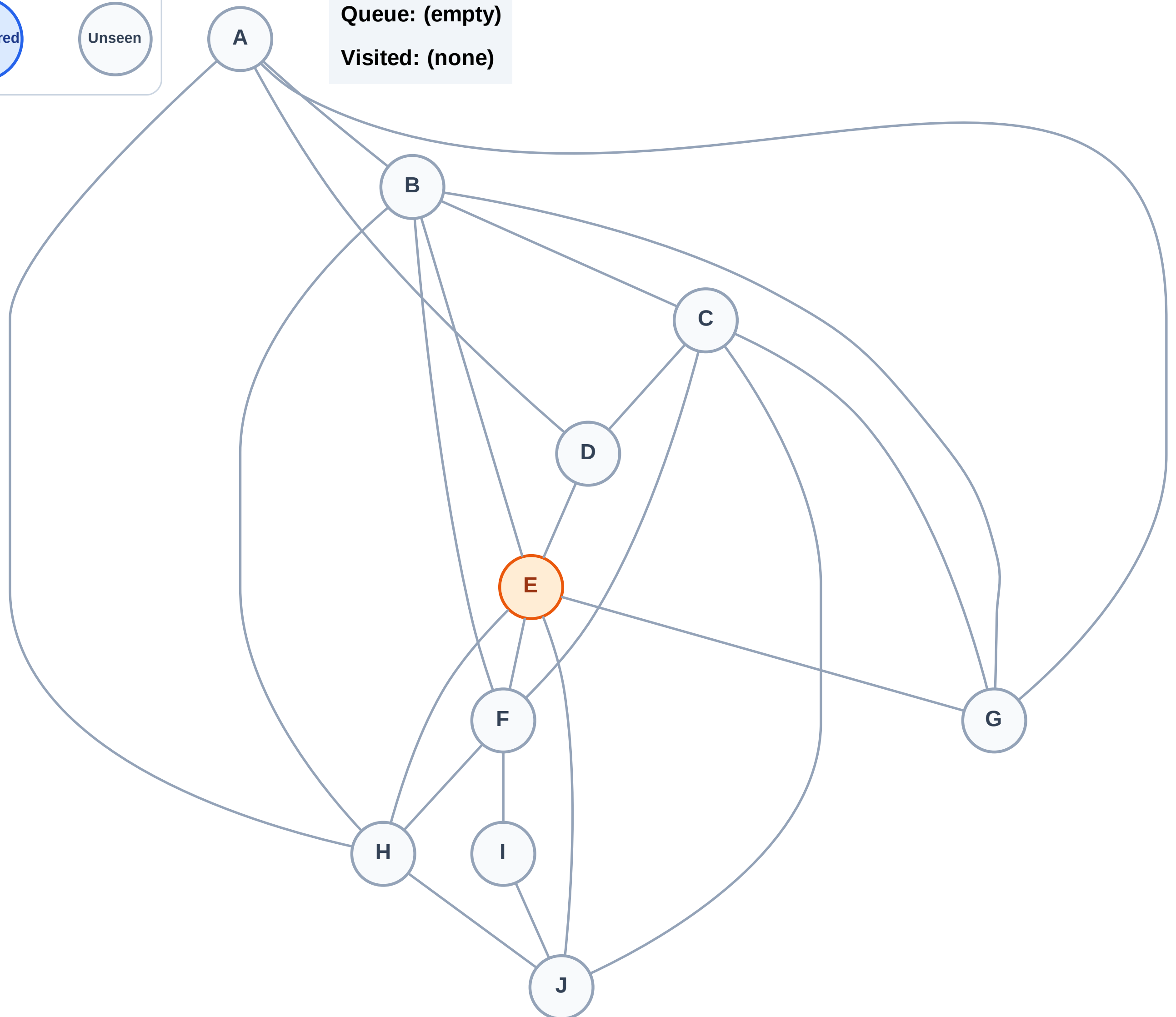
Step 2: Process node E

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

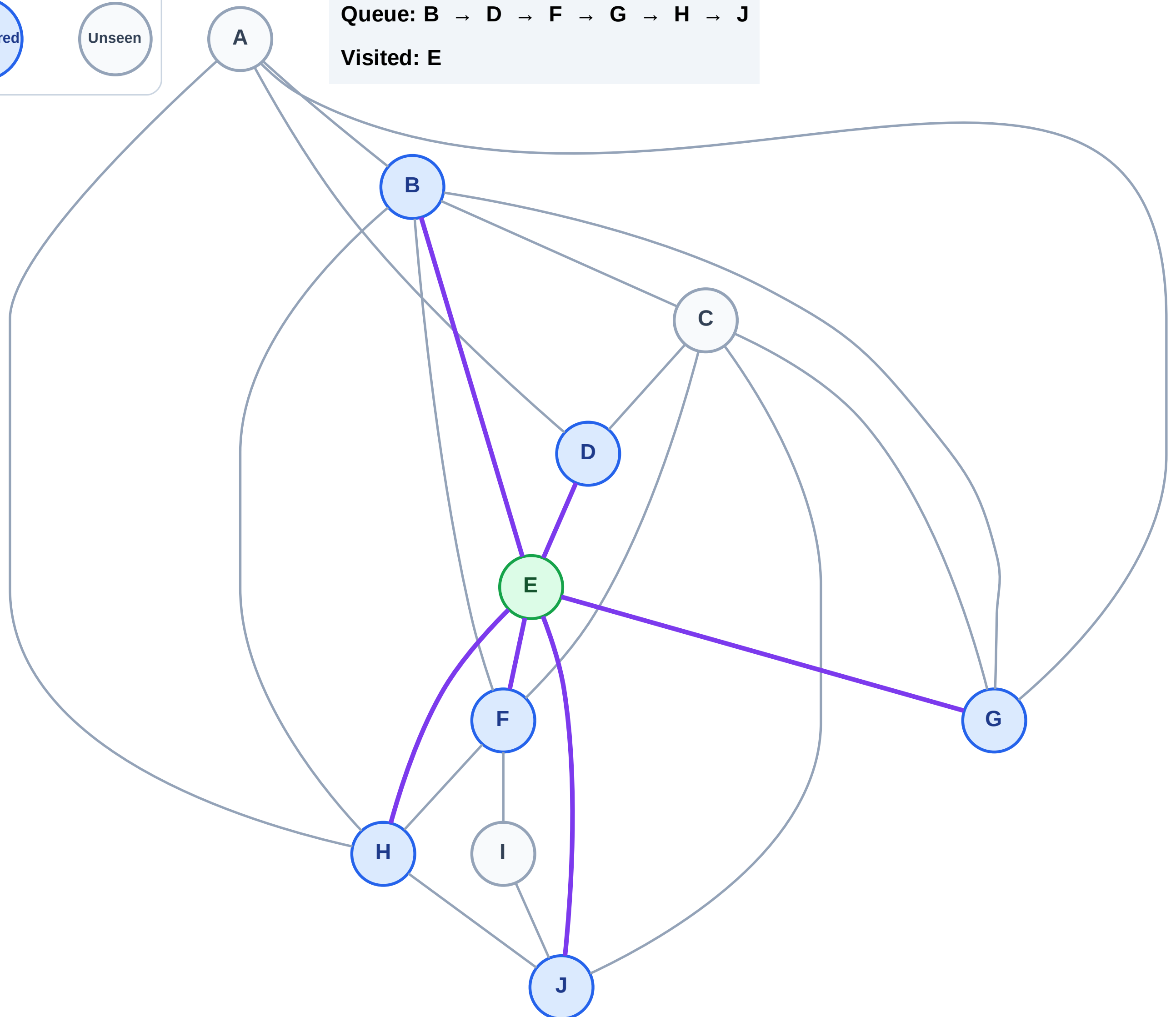
Step 3: Discover B, D, F, G, H, J from E

Legend



Queue: B → D → F → G → H → J

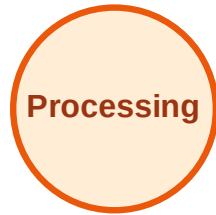
Visited: E



BFS Traversal

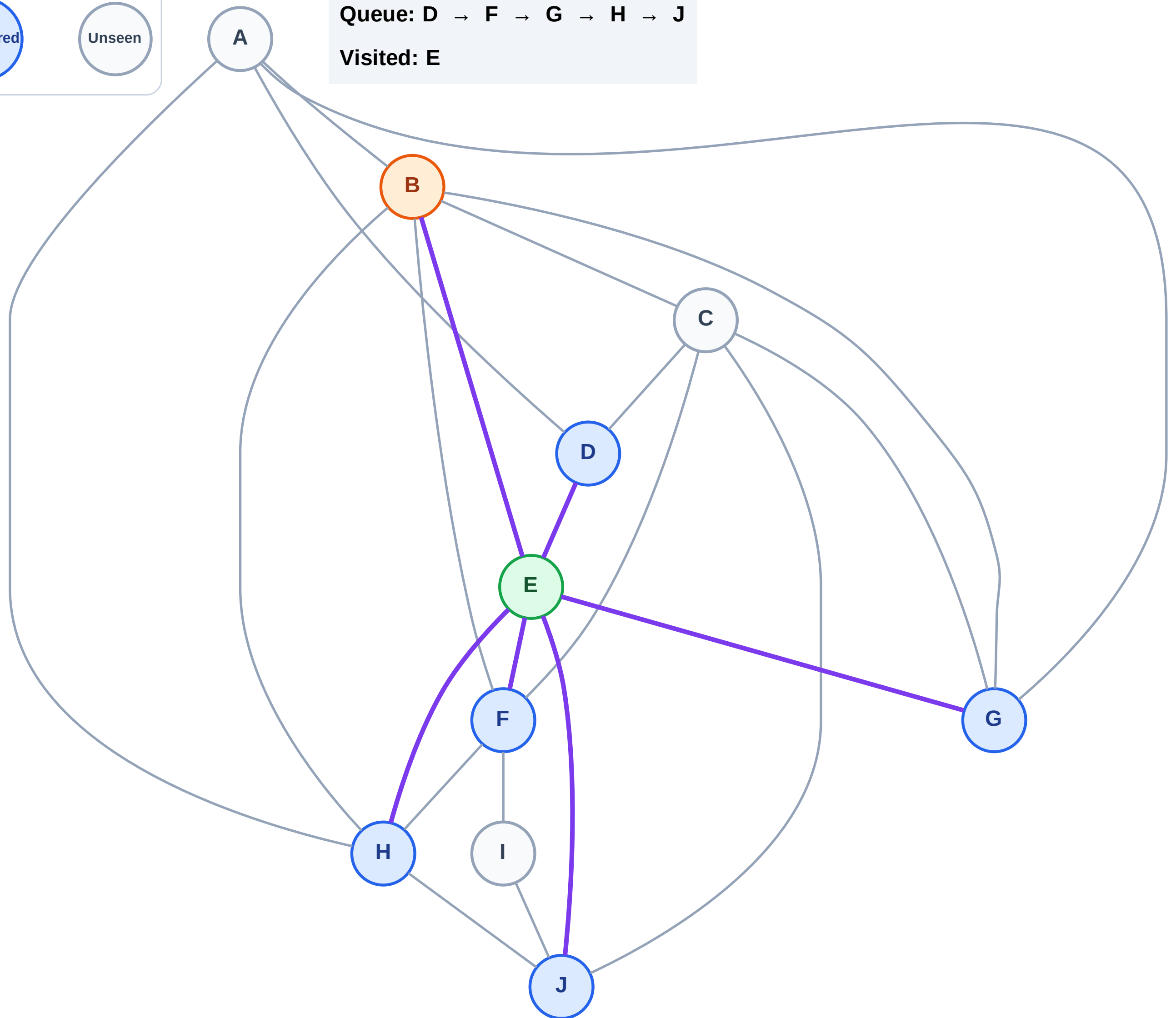
Step 4: Process node B

Legend



Queue: D → F → G → H → J

Visited: E



BFS Traversal

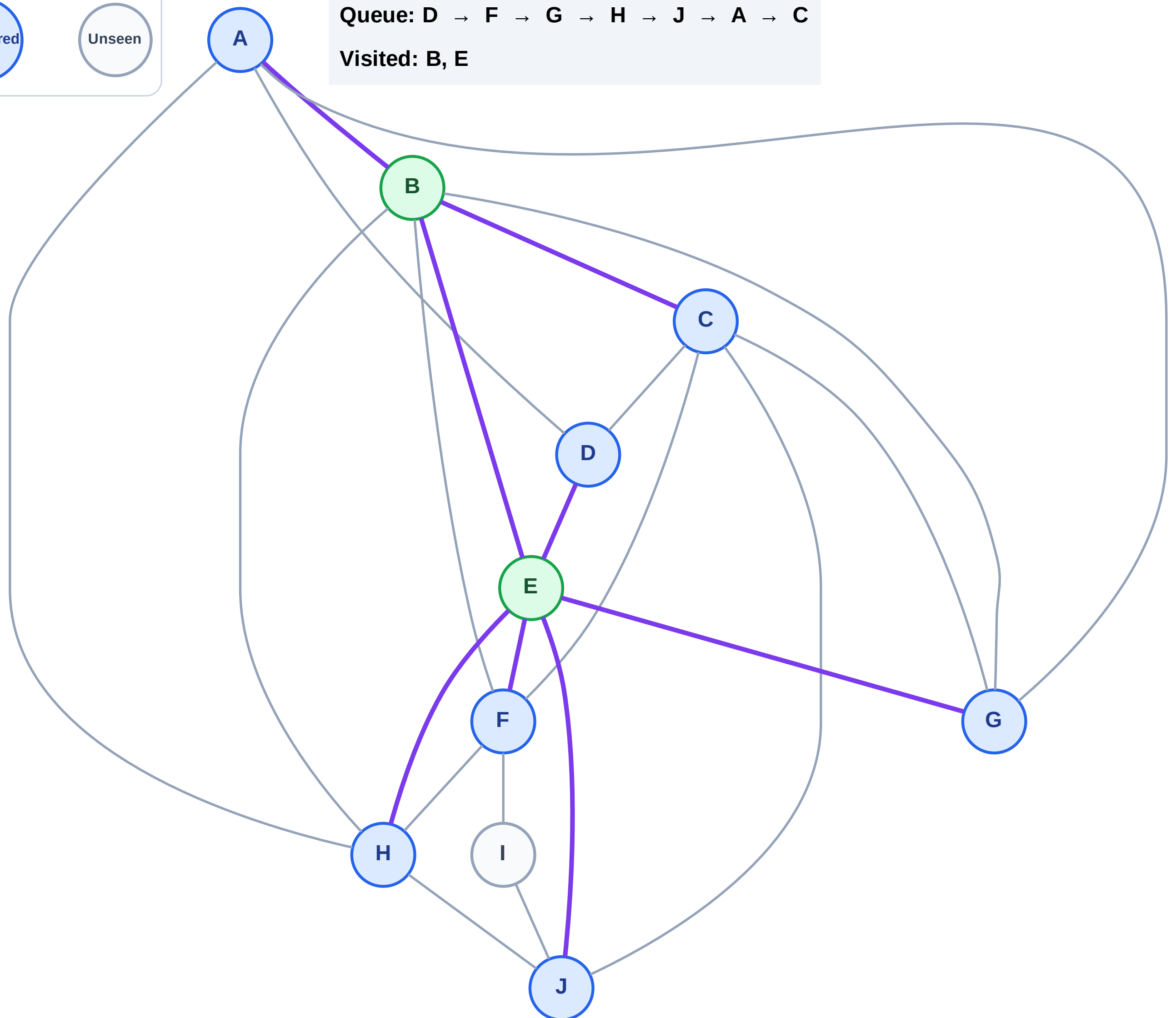
Step 5: Discover A, C from B

Legend



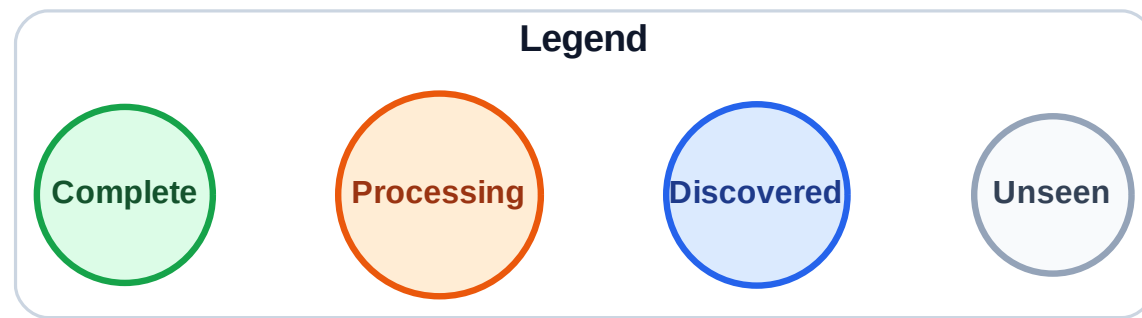
Queue: D → F → G → H → J → A → C

Visited: B, E

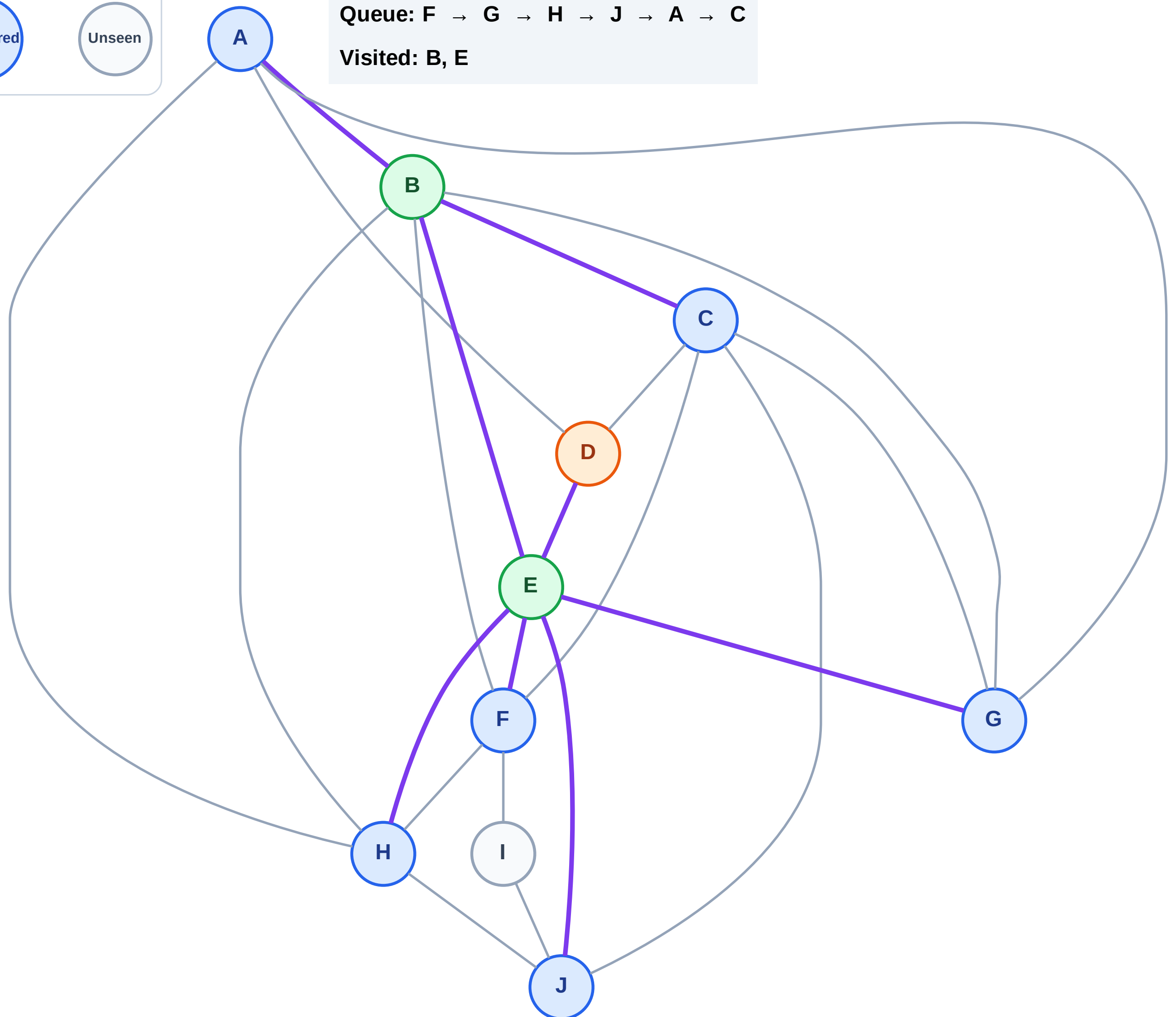


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Step 6: Process node D



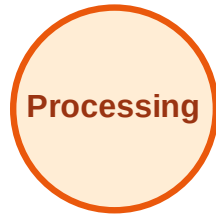
Queue: F → G → H → J → A → C
Visited: B, E



BFS Traversal

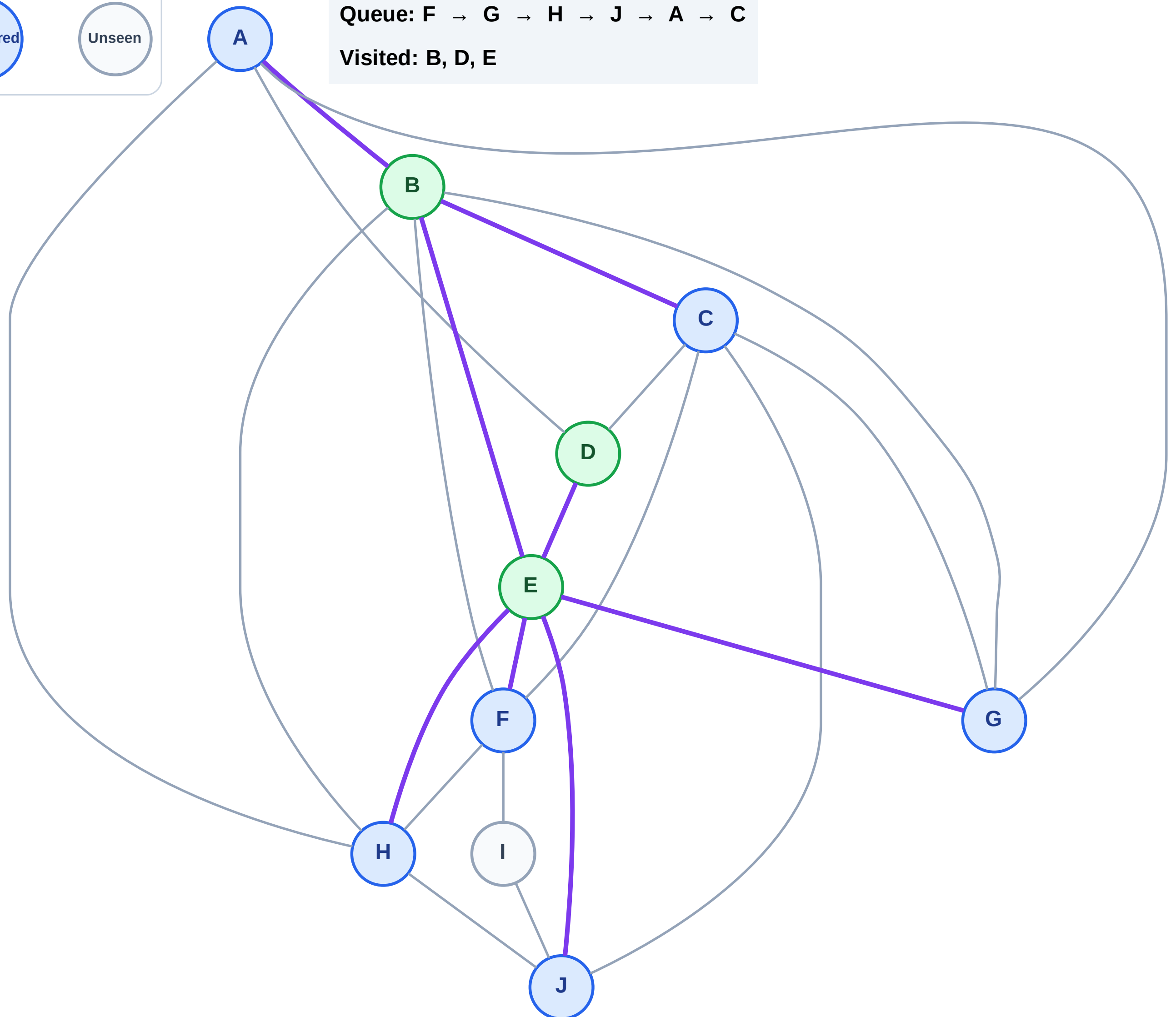
Step 7: No new neighbors from D

Legend



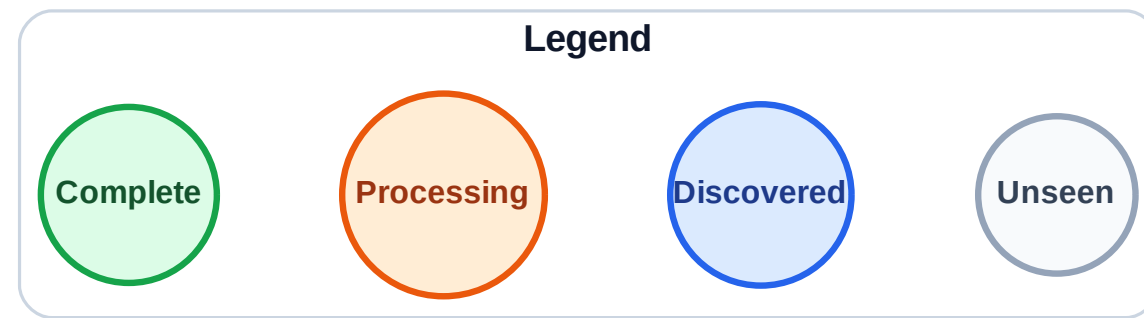
Queue: F → G → H → J → A → C

Visited: B, D, E



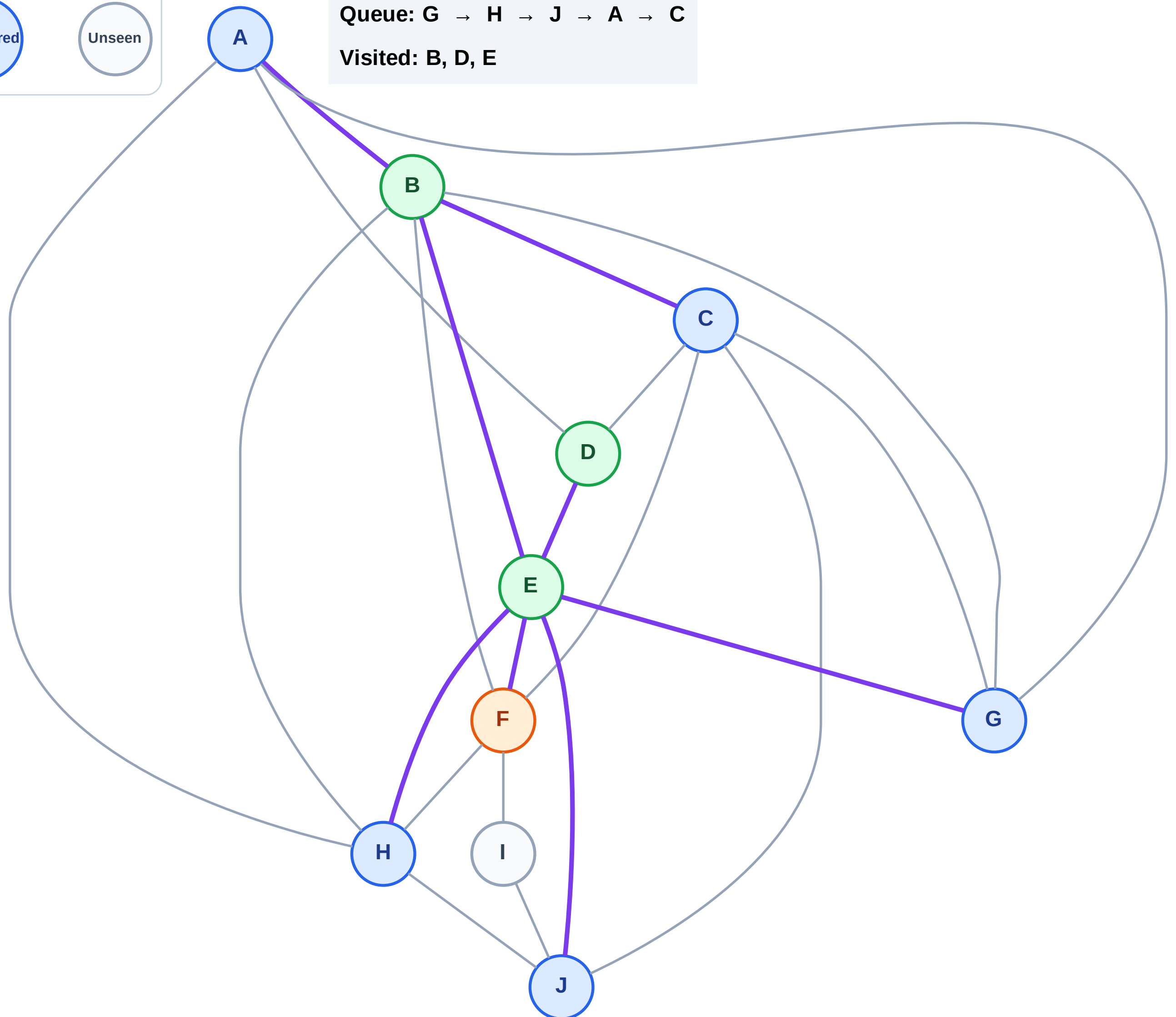
BFS Traversal

Step 8: Process node F



Queue: G → H → J → A → C

Visited: B, D, E



BFS Traversal

Step 9: Discover I from F

Legend

Complete

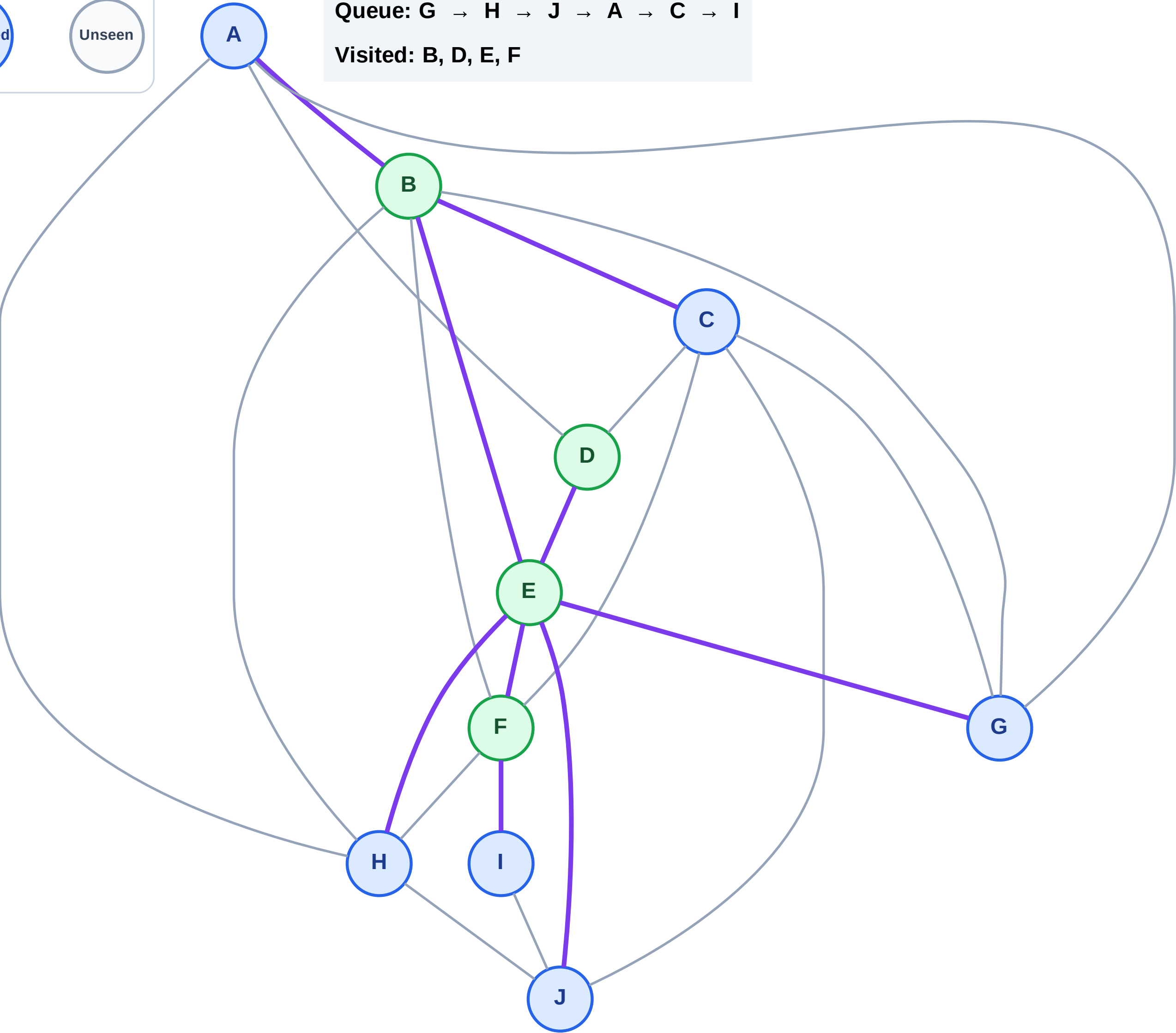
Processing

Discovered

Unseen

Queue: G → H → J → A → C → I

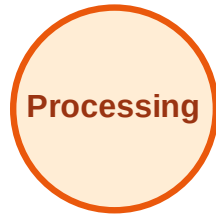
Visited: B, D, E, F



BFS Traversal

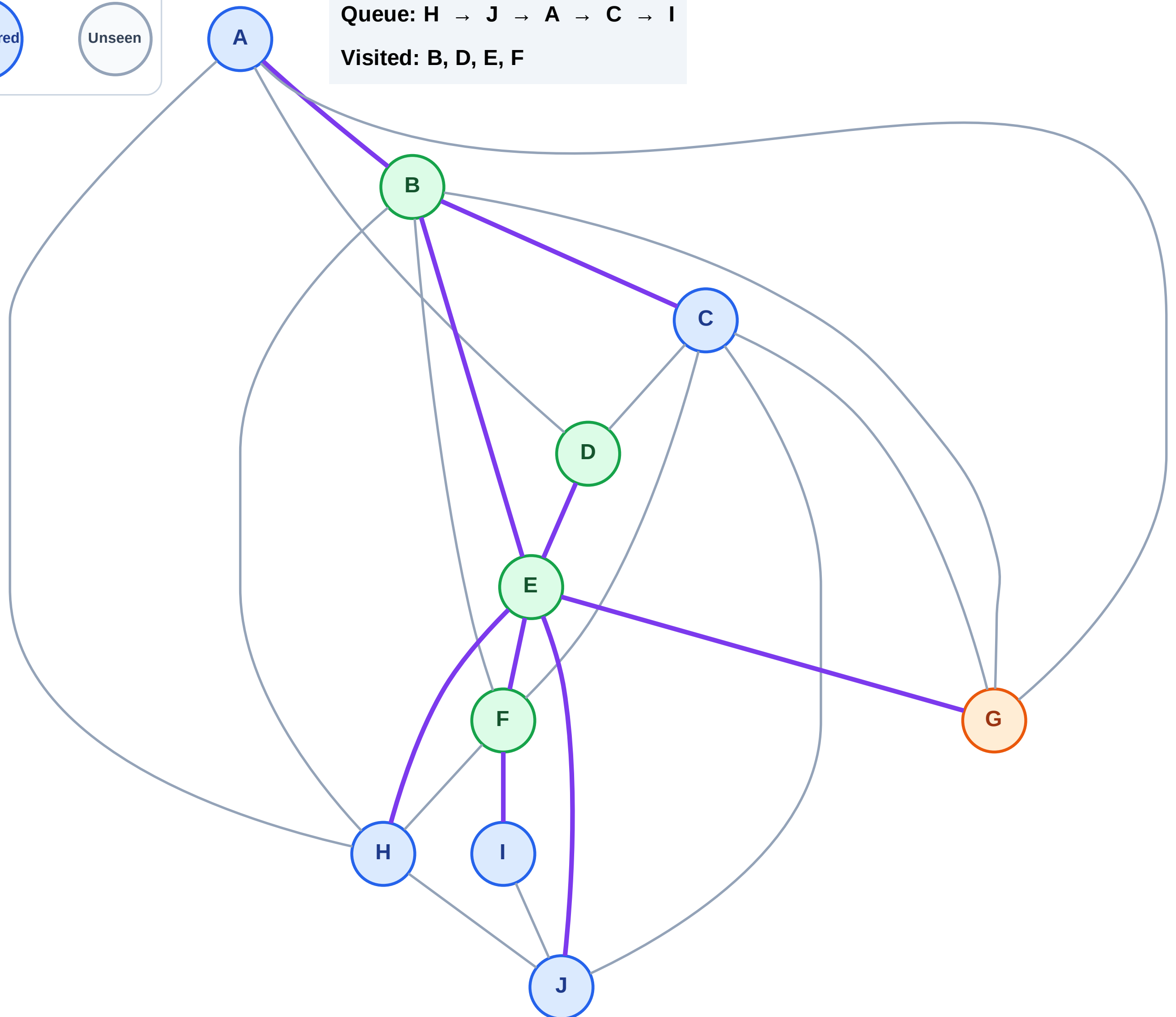
Step 10: Process node G

Legend



Queue: H → J → A → C → I

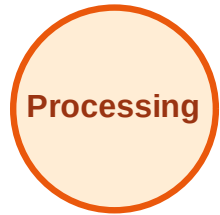
Visited: B, D, E, F



BFS Traversal

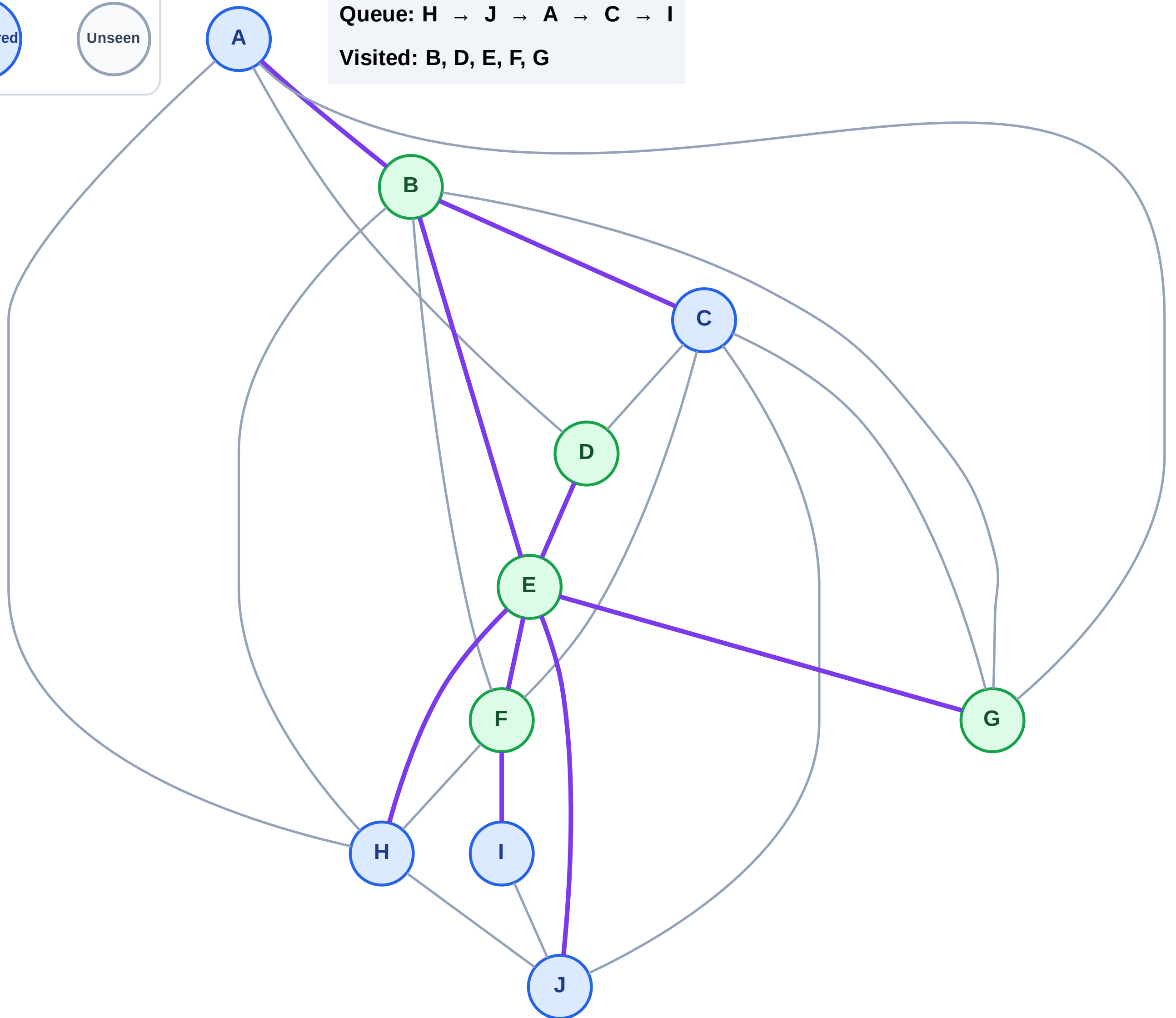
Step 11: No new neighbors from G

Legend



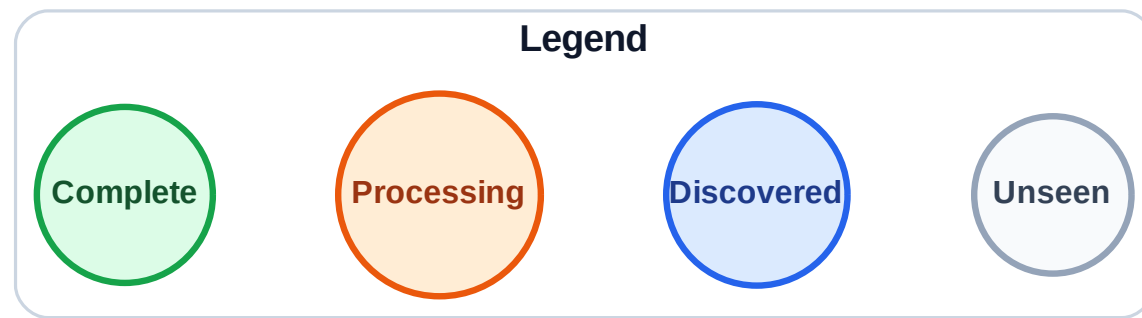
Queue: H → J → A → C → I

Visited: B, D, E, F, G

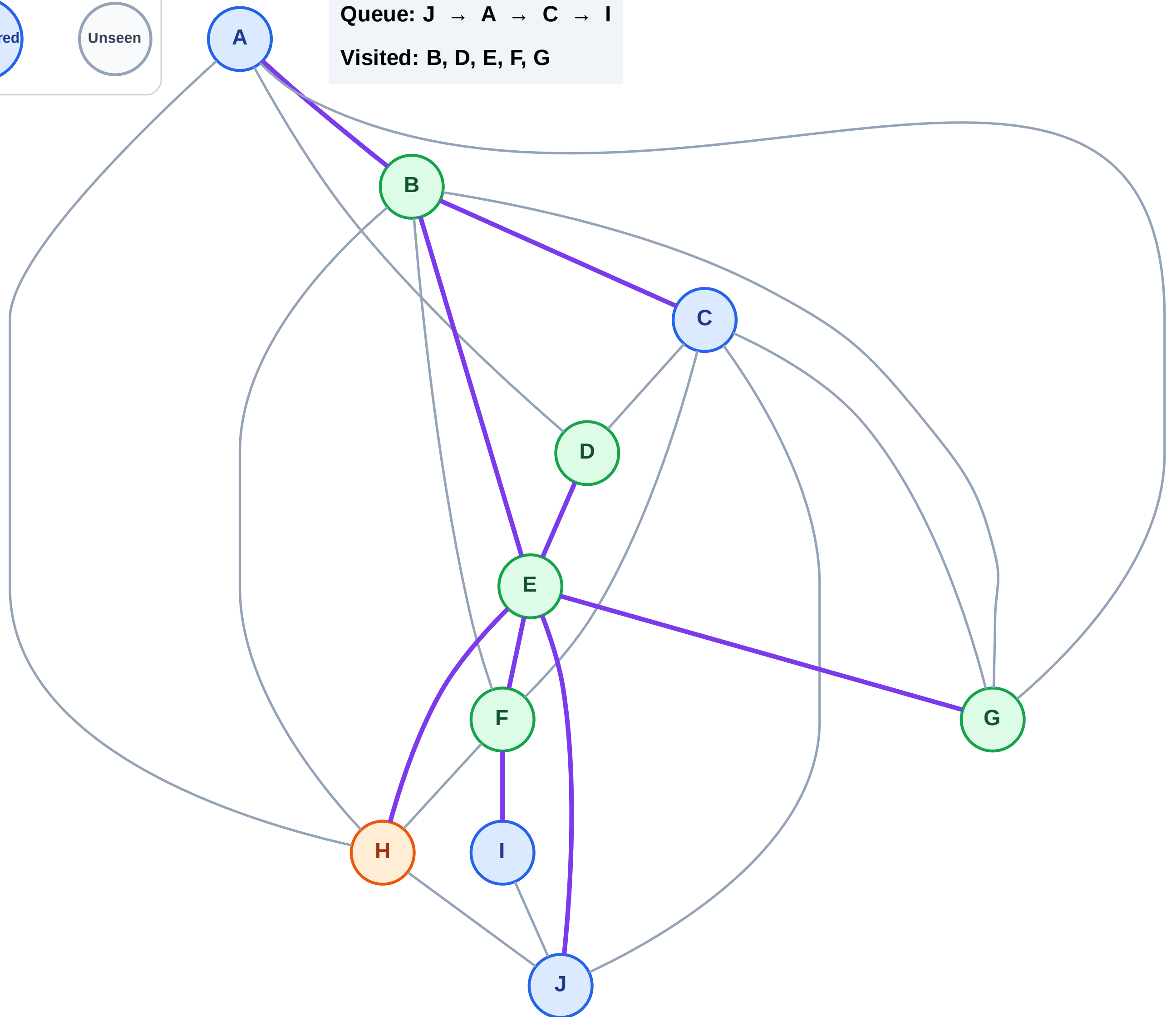


BFS Traversal

Step 12: Process node H



Queue: J → A → C → I
Visited: B, D, E, F, G



BFS Traversal

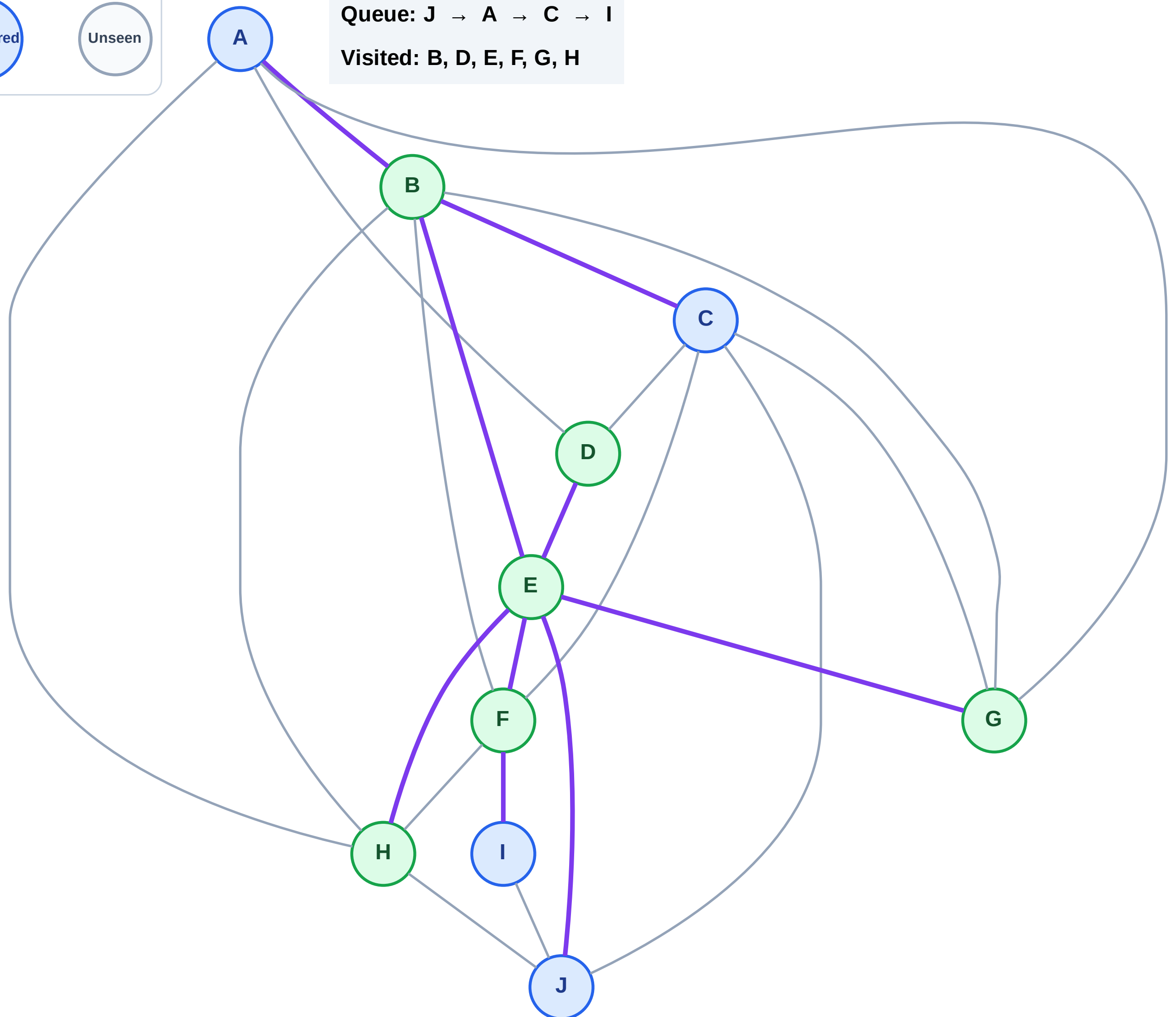
Step 13: No new neighbors from H

Legend



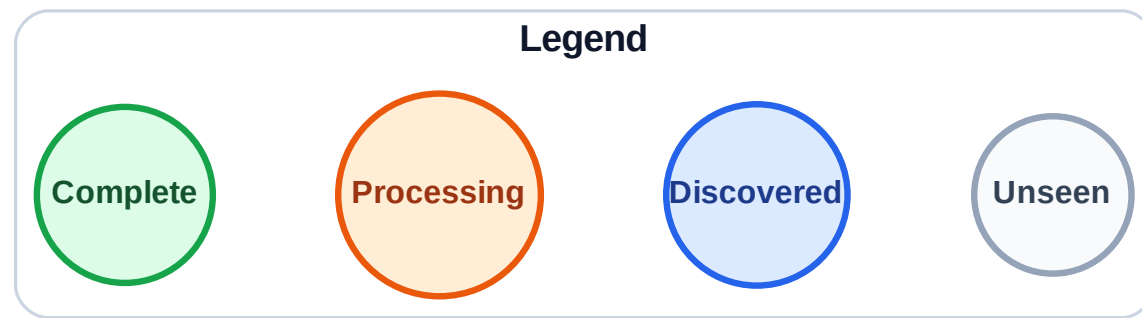
Queue: J → A → C → I

Visited: B, D, E, F, G, H

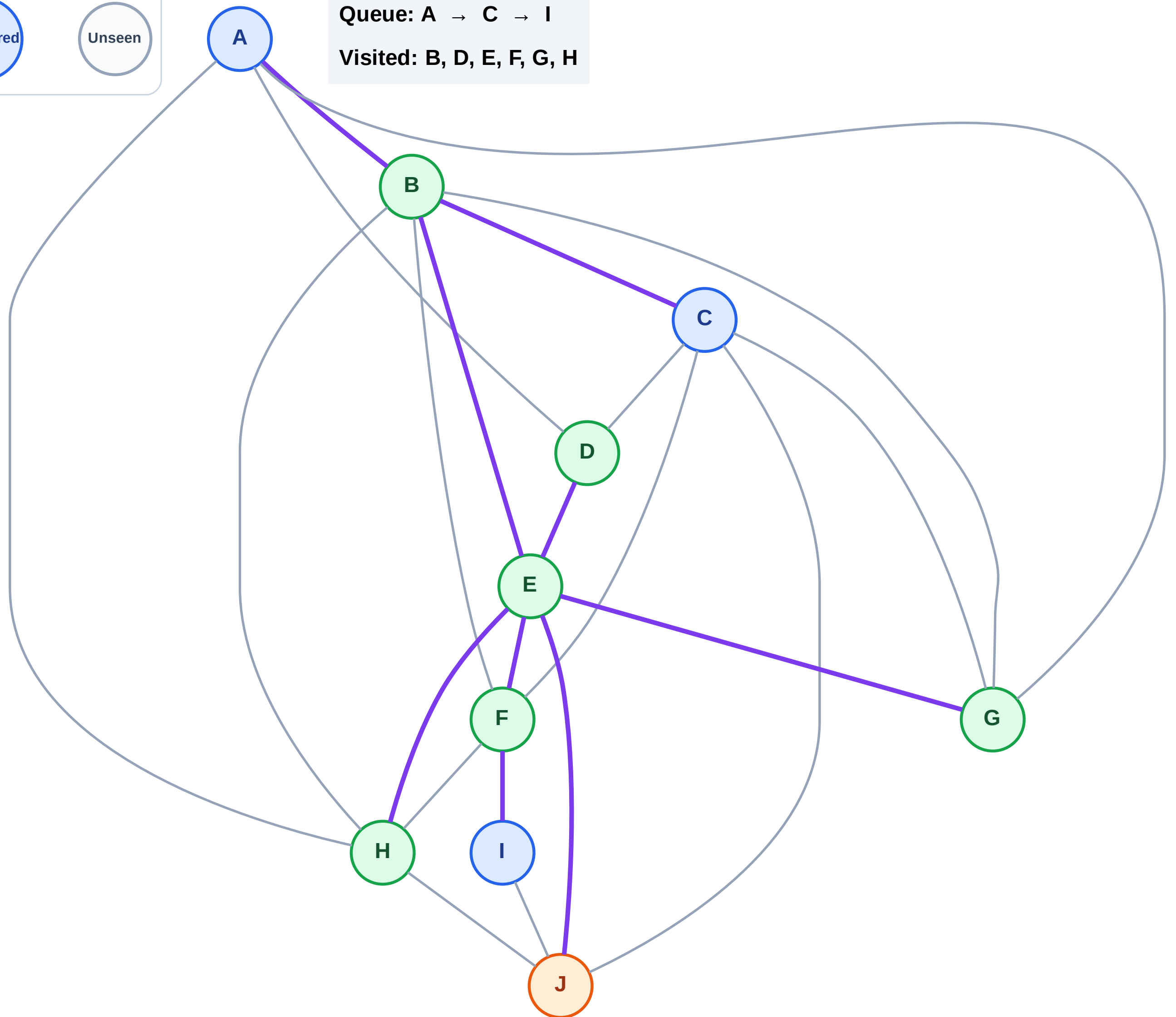


BFS Traversal

Step 14: Process node J



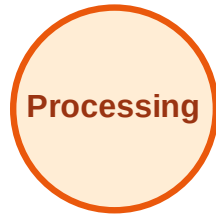
Queue: A → C → I
Visited: B, D, E, F, G, H



BFS Traversal

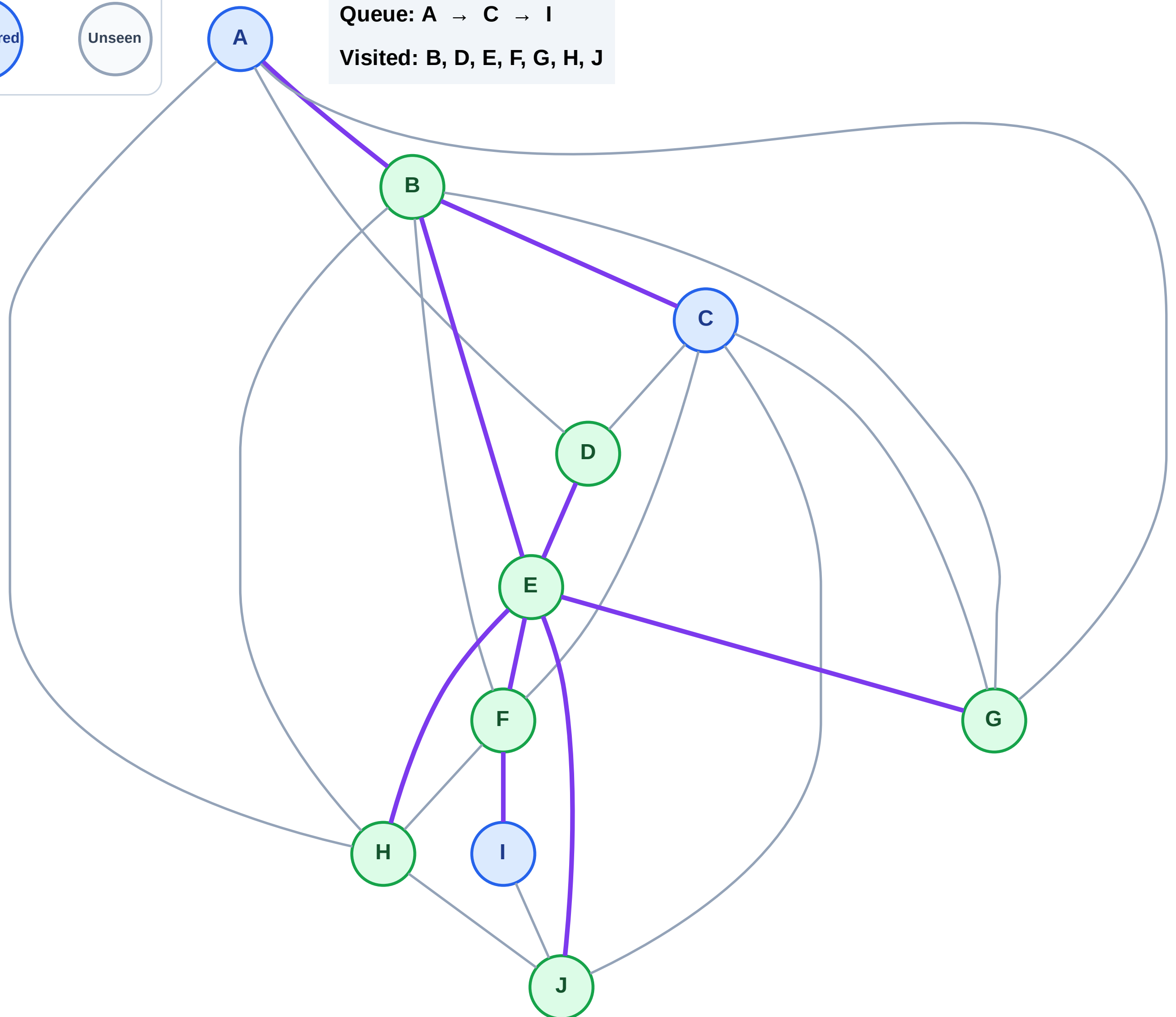
Step 15: No new neighbors from J

Legend



Queue: A → C → I

Visited: B, D, E, F, G, H, J



BFS Traversal

Step 16: Process node A

Legend

Complete

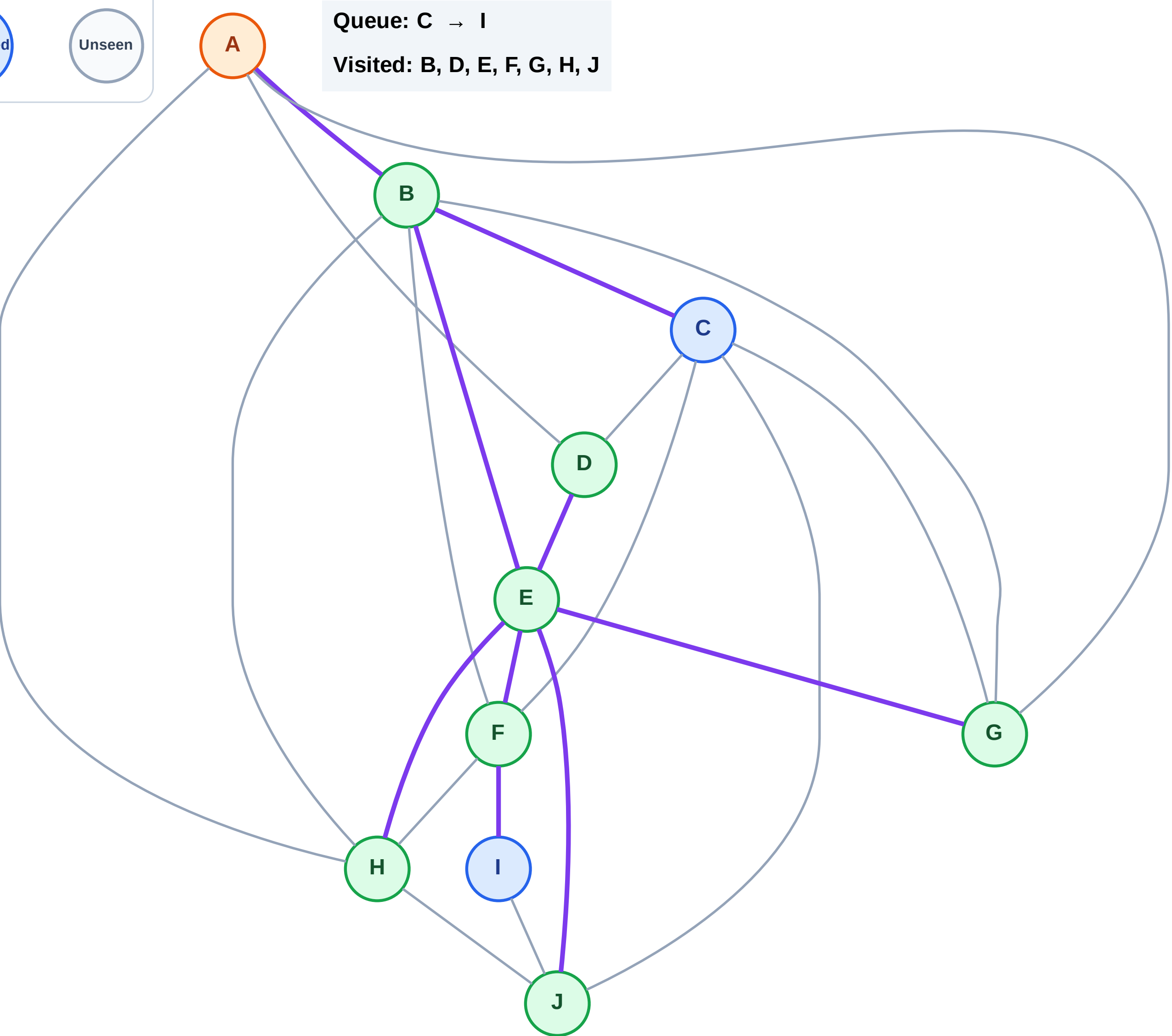
Processing

Discovered

Unseen

Queue: C → I

Visited: B, D, E, F, G, H, J



BFS Traversal

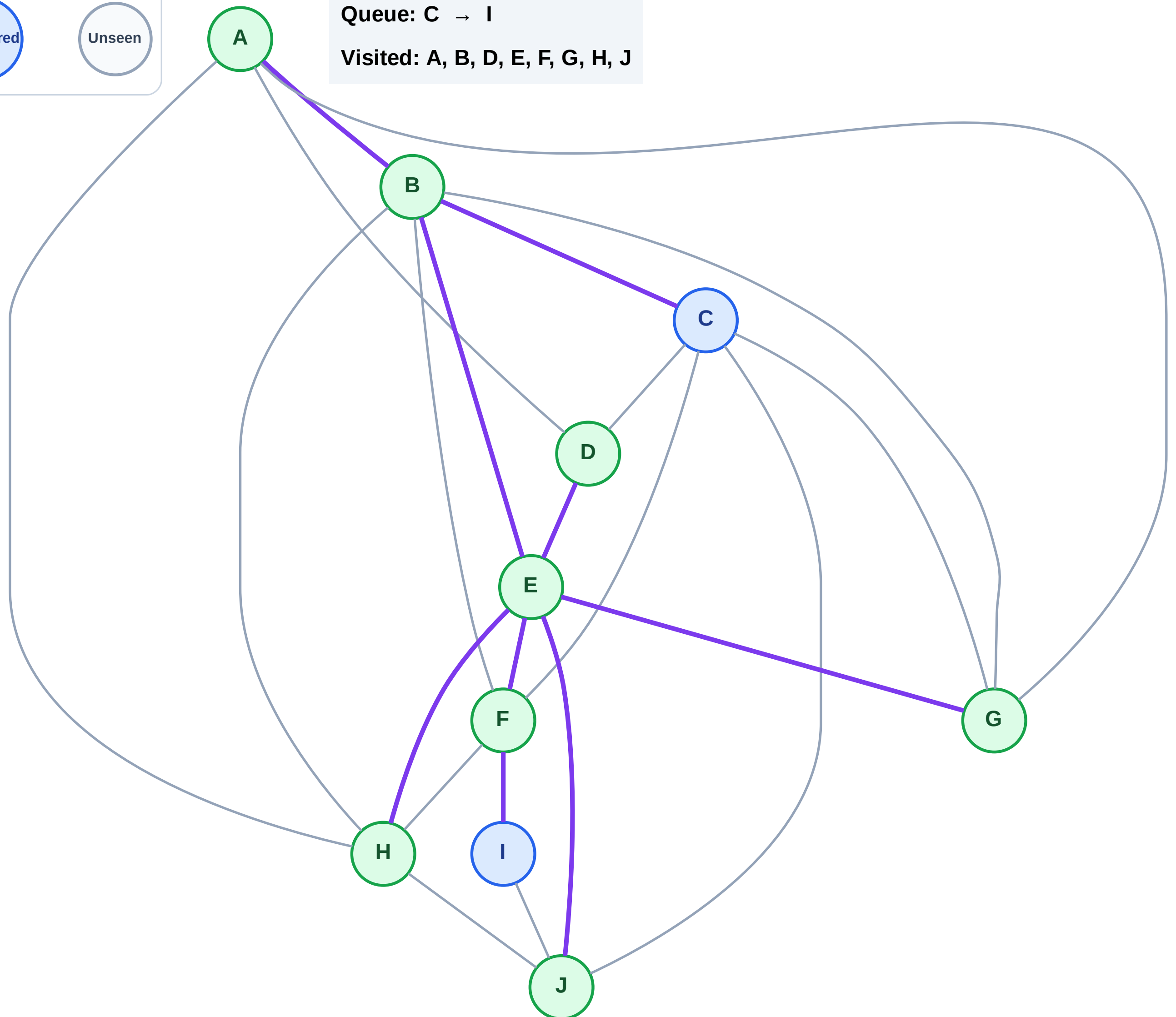
Step 17: No new neighbors from A

Legend



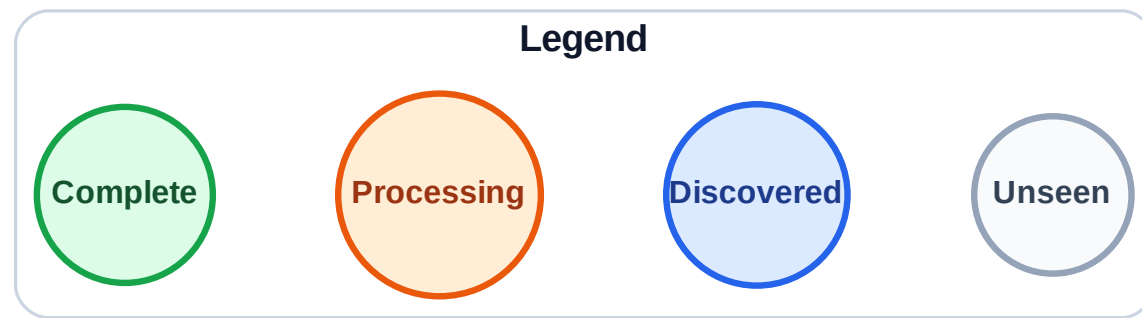
Queue: C → I

Visited: A, B, D, E, F, G, H, J

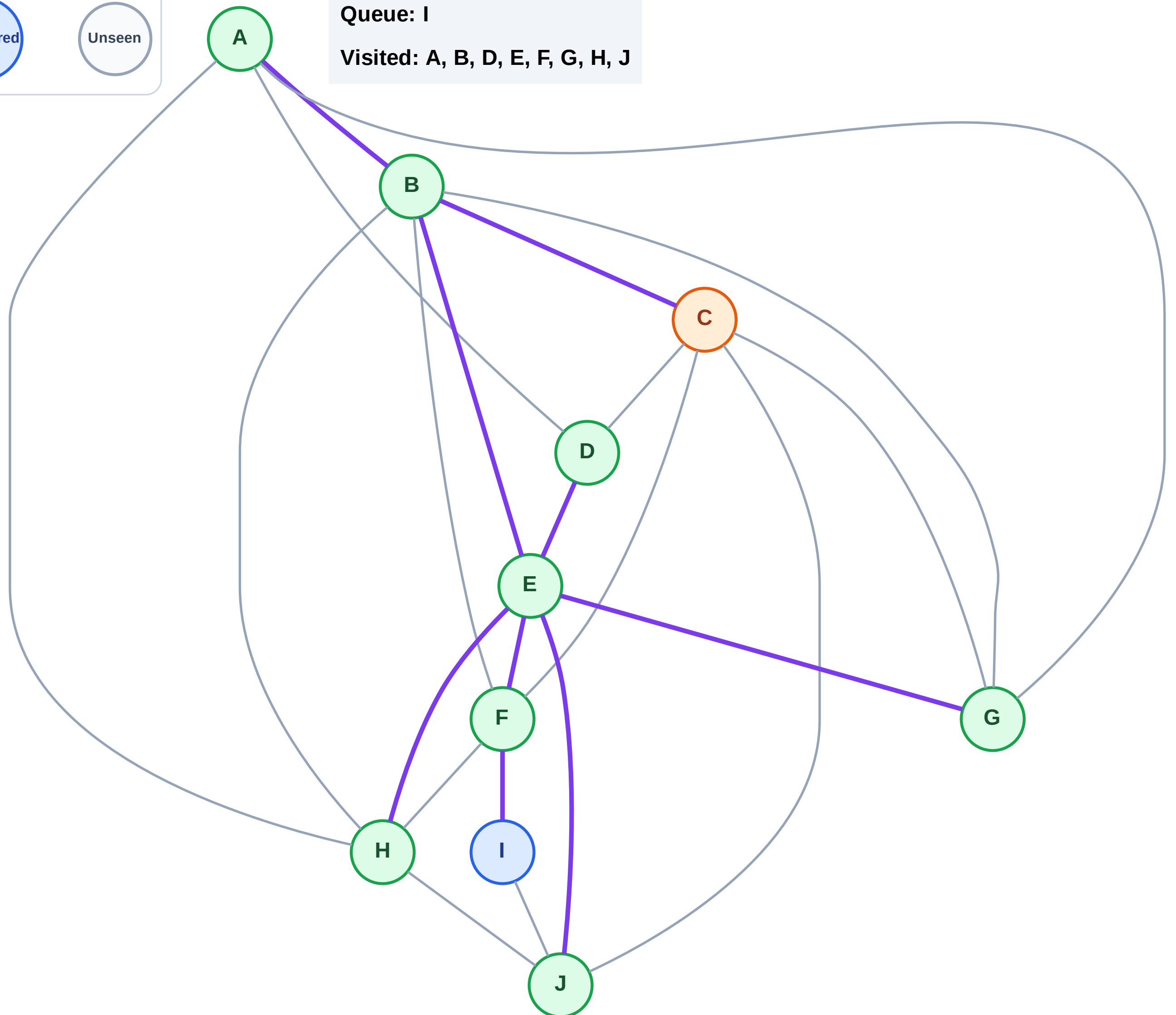


BFS Traversal

Step 18: Process node C



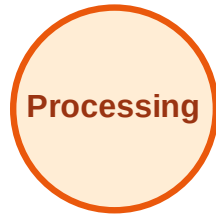
Queue: I
Visited: A, B, D, E, F, G, H, J



BFS Traversal

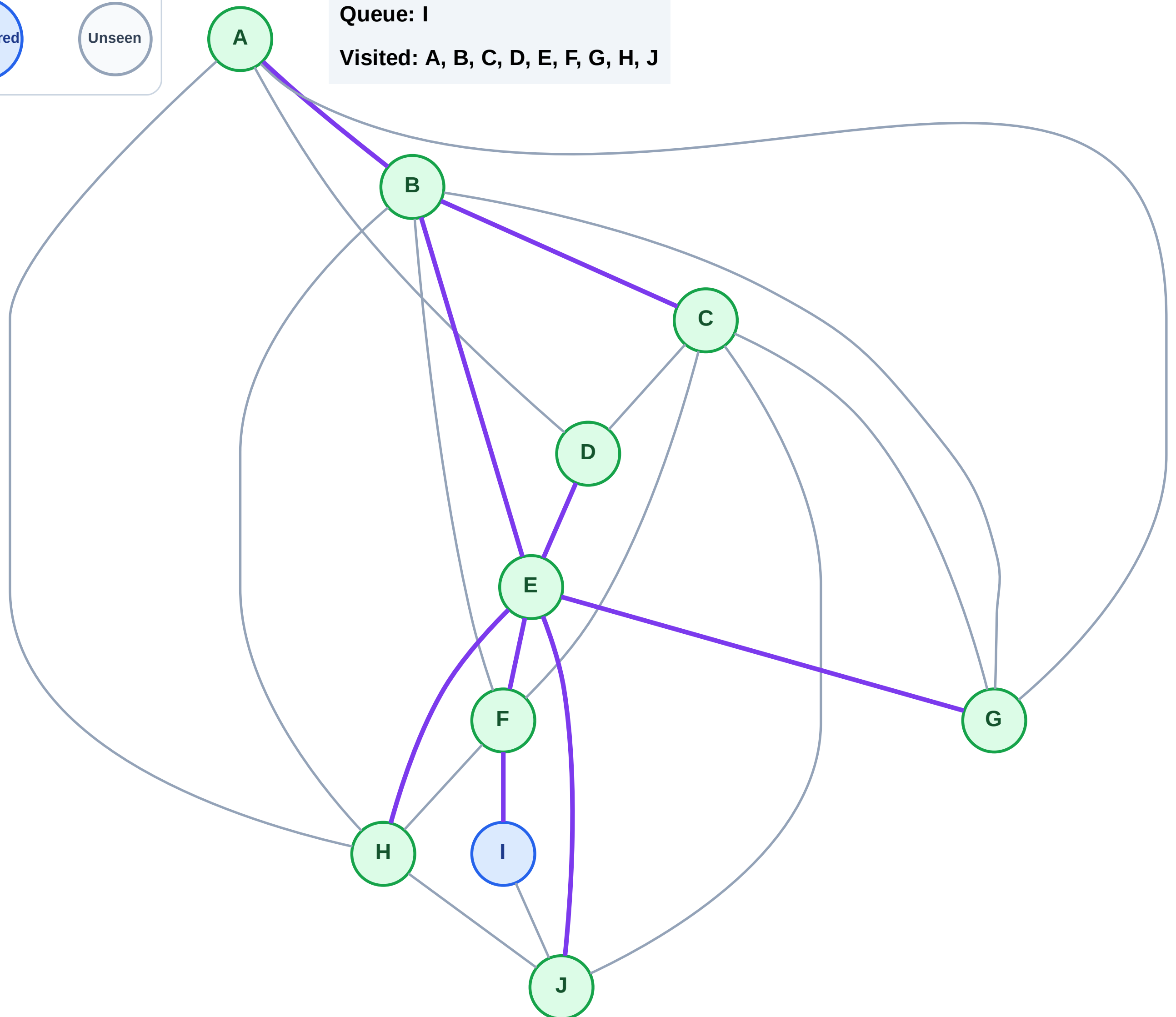
Step 19: No new neighbors from C

Legend



Queue: I

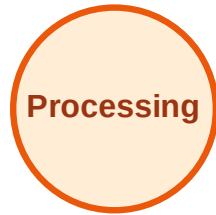
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

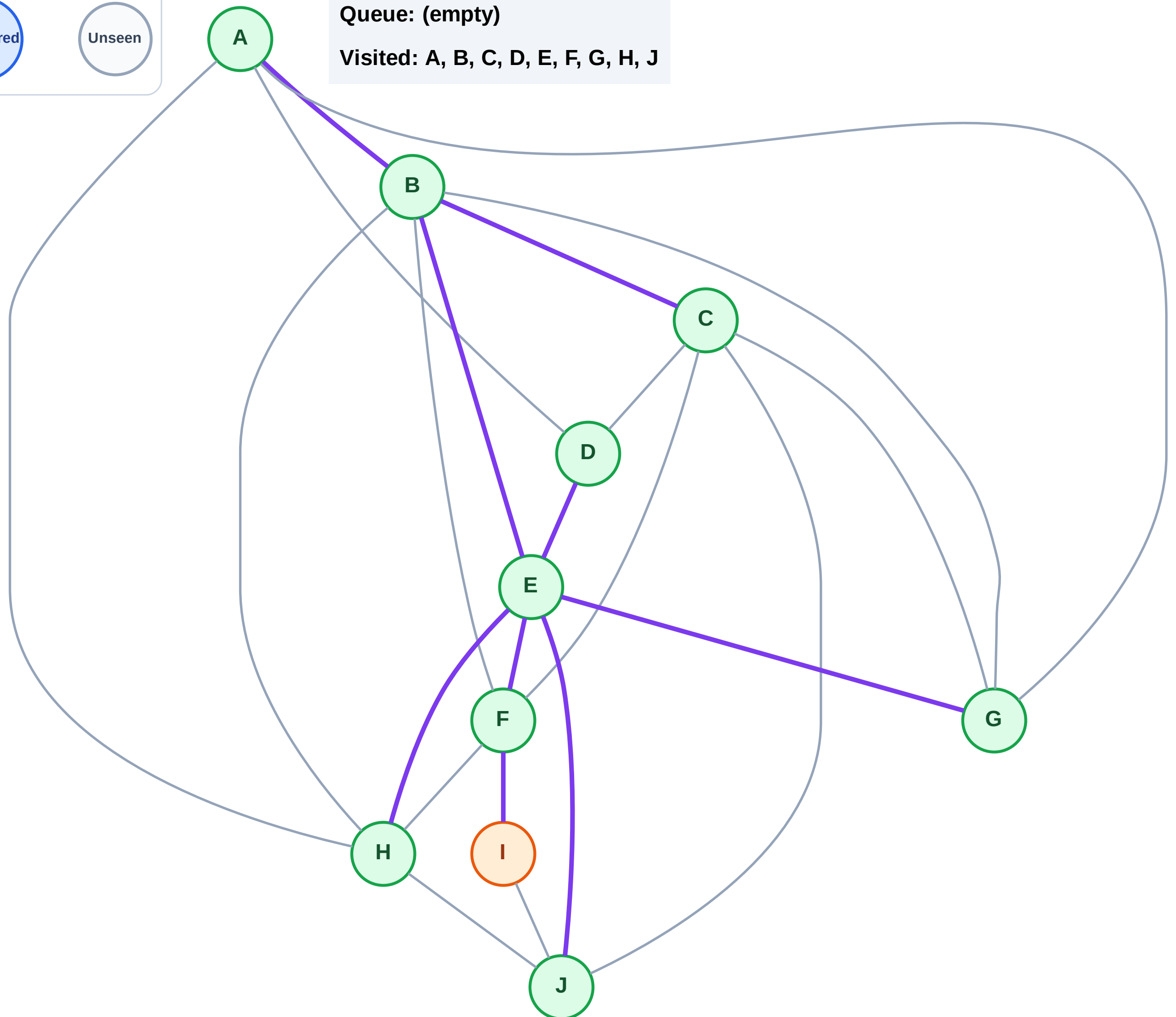
Step 20: Process node I

Legend



Queue: (empty)

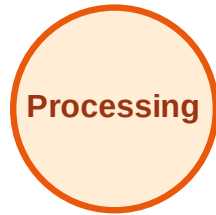
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

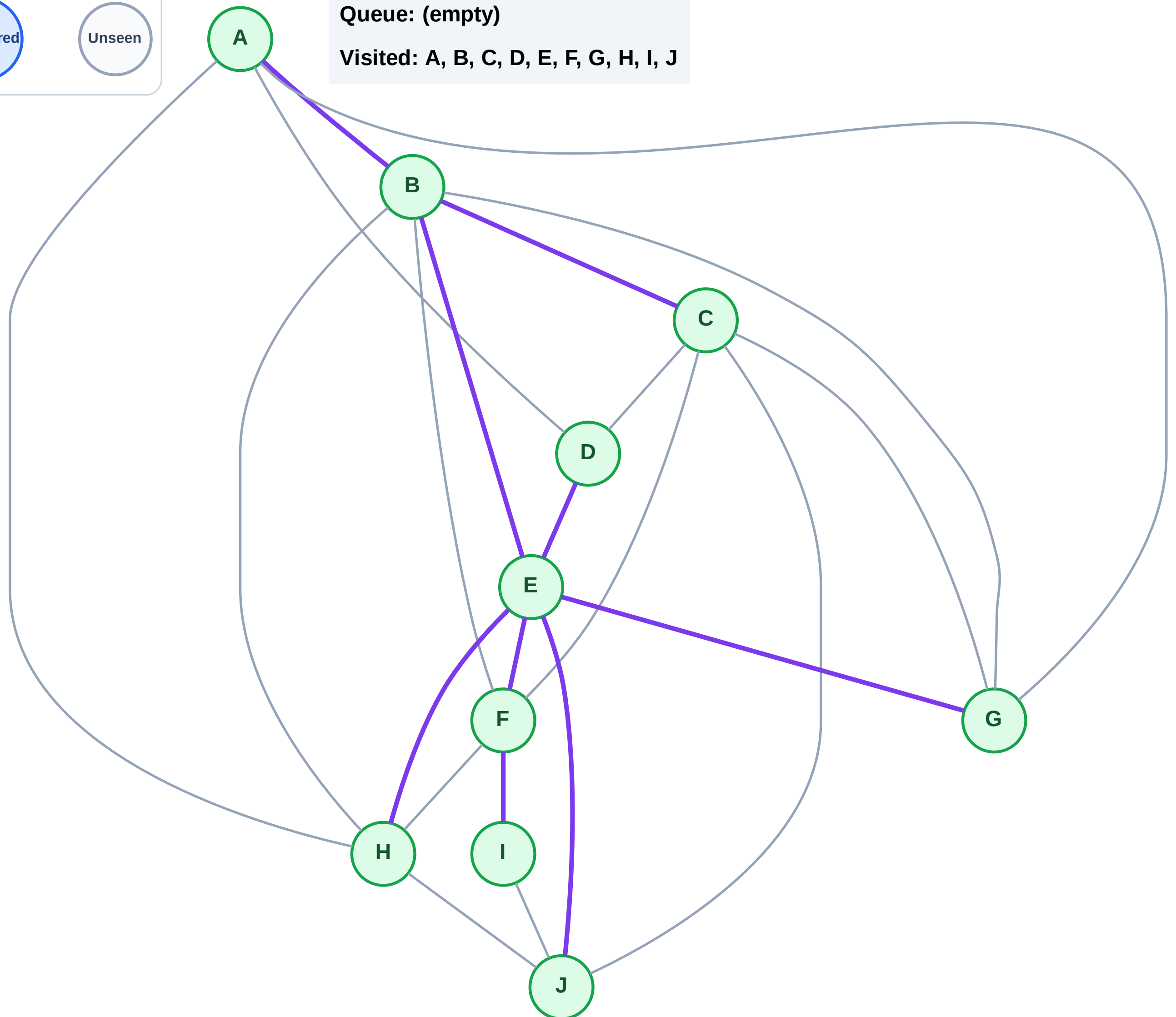
Step 21: No new neighbors from I

Legend



Queue: (empty)

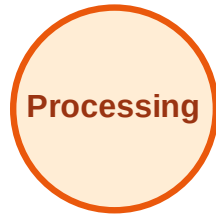
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

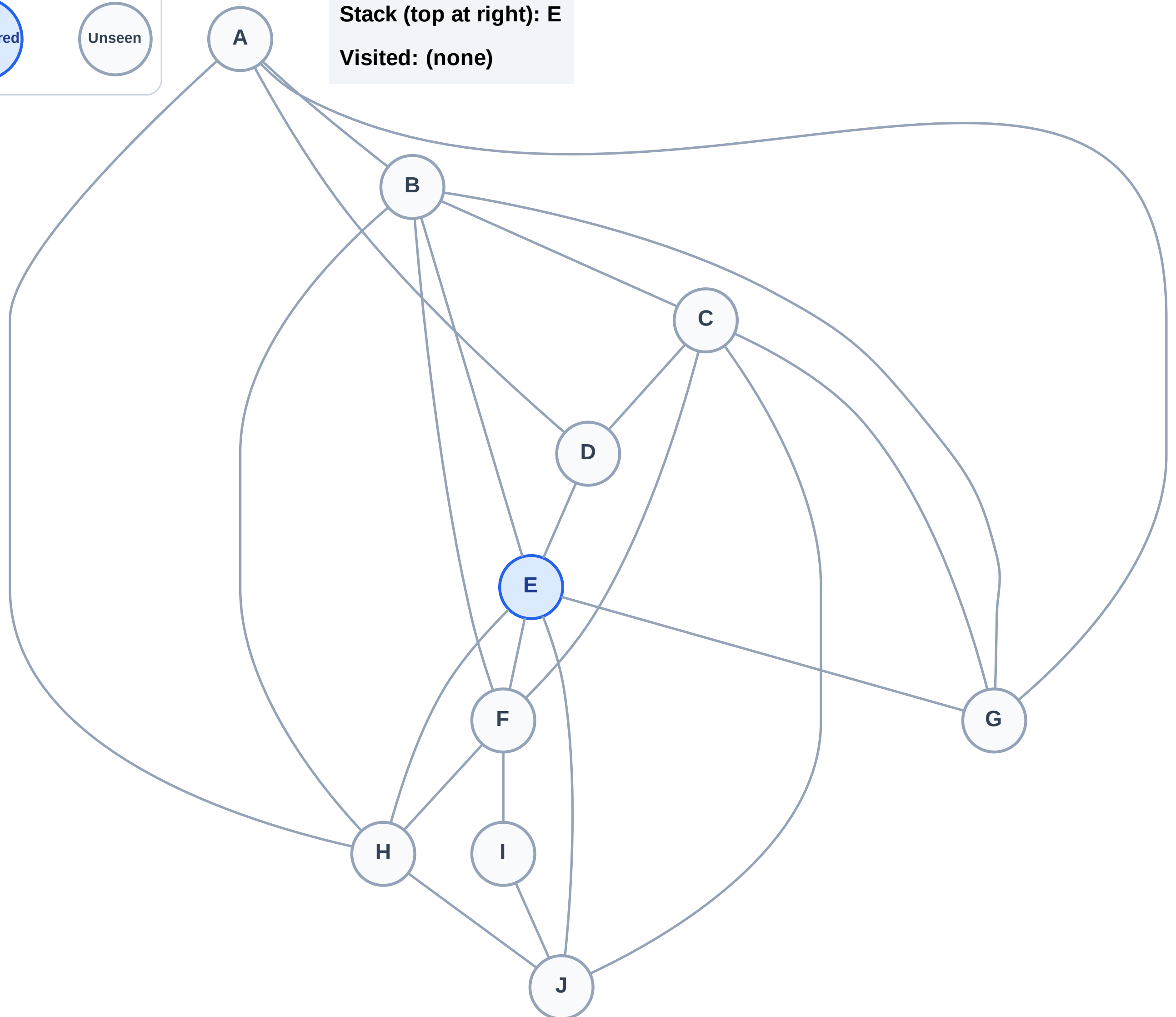
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

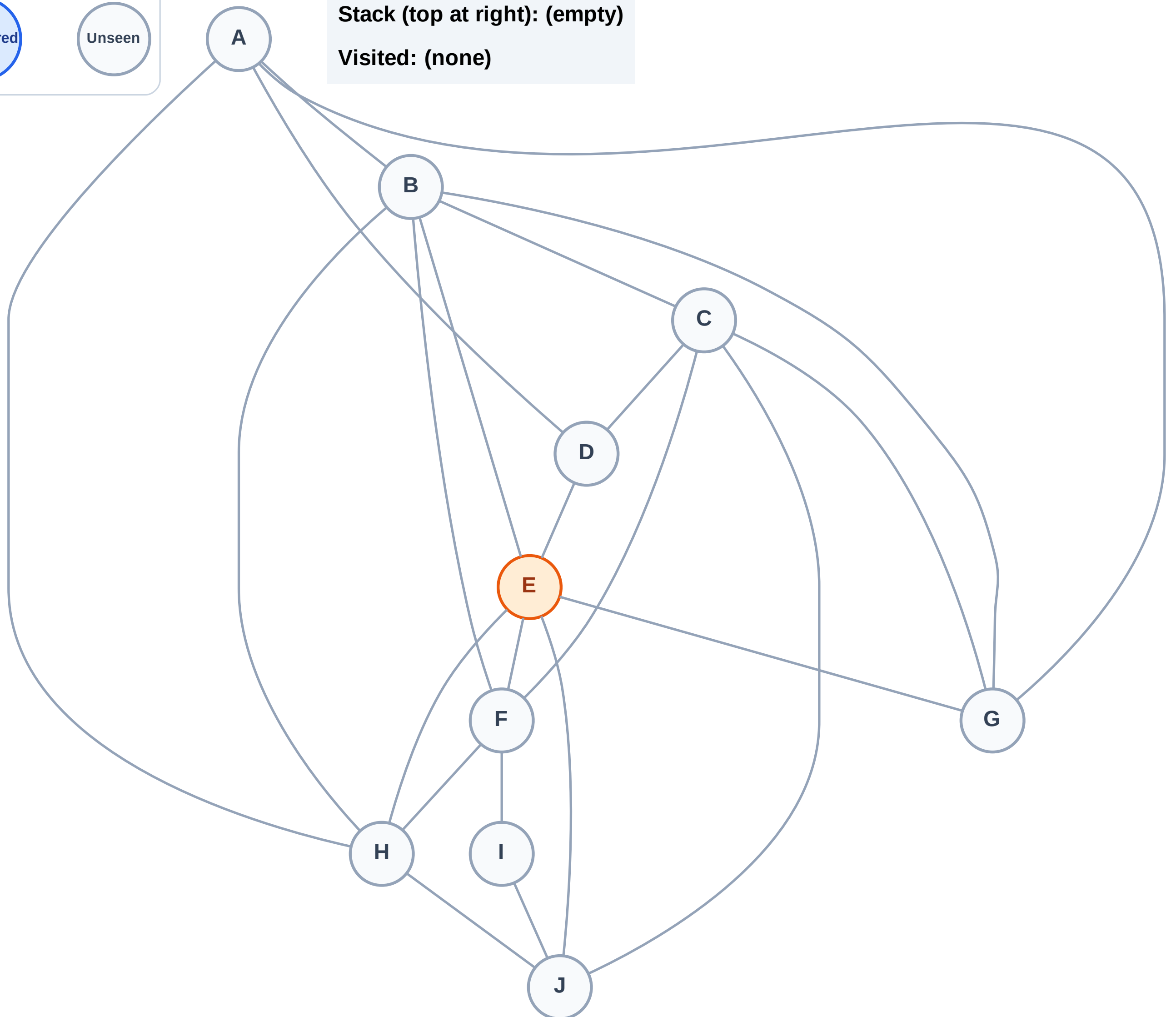
Step 2: Process node E

Legend



Stack (top at right): (empty)

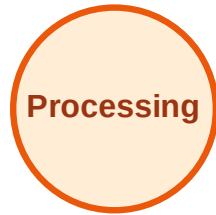
Visited: (none)



DFS Traversal

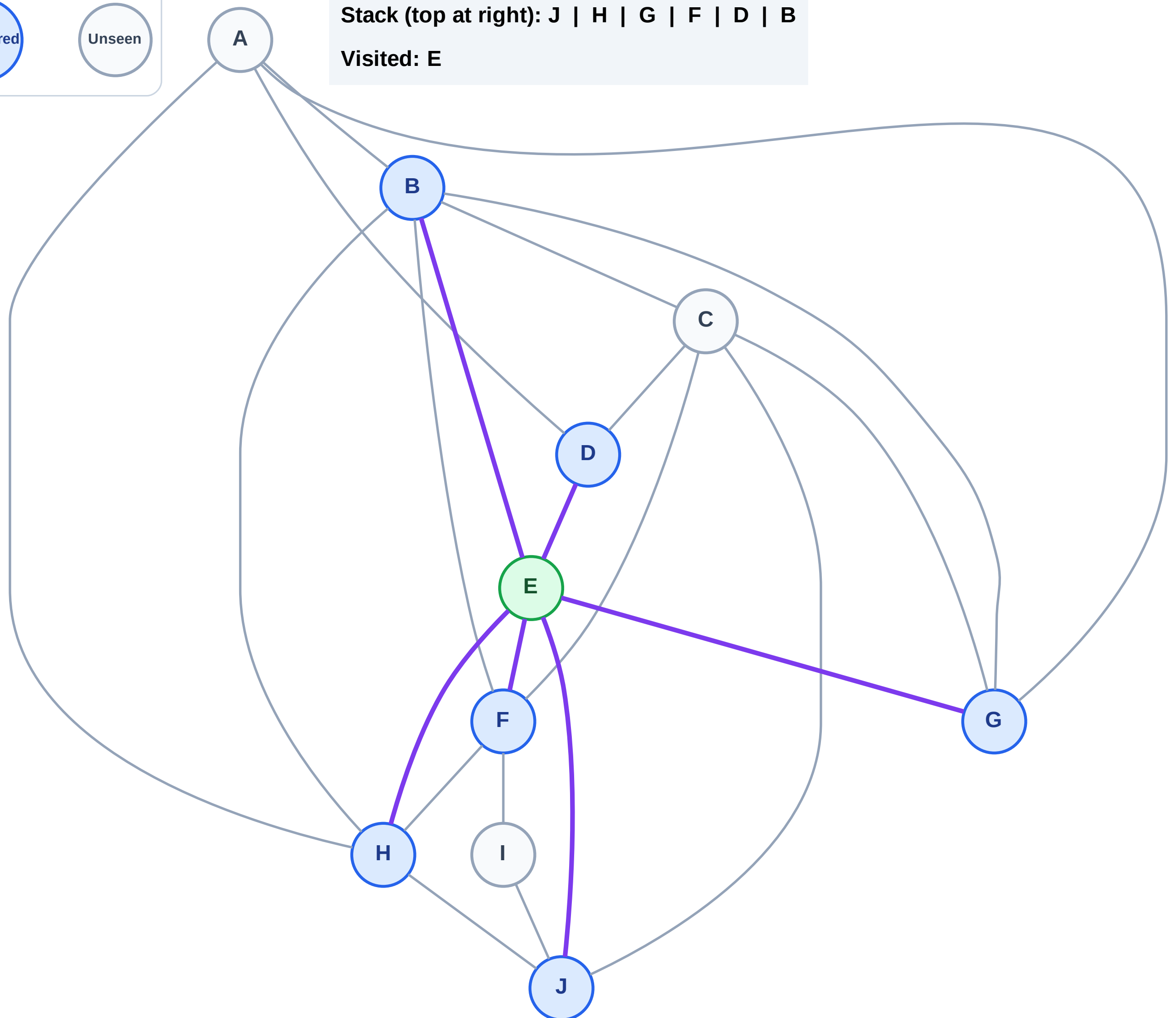
Step 3: Discover J, H, G, F, D, B from E

Legend



Stack (top at right): J | H | G | F | D | B

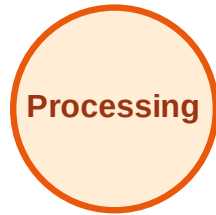
Visited: E



DFS Traversal

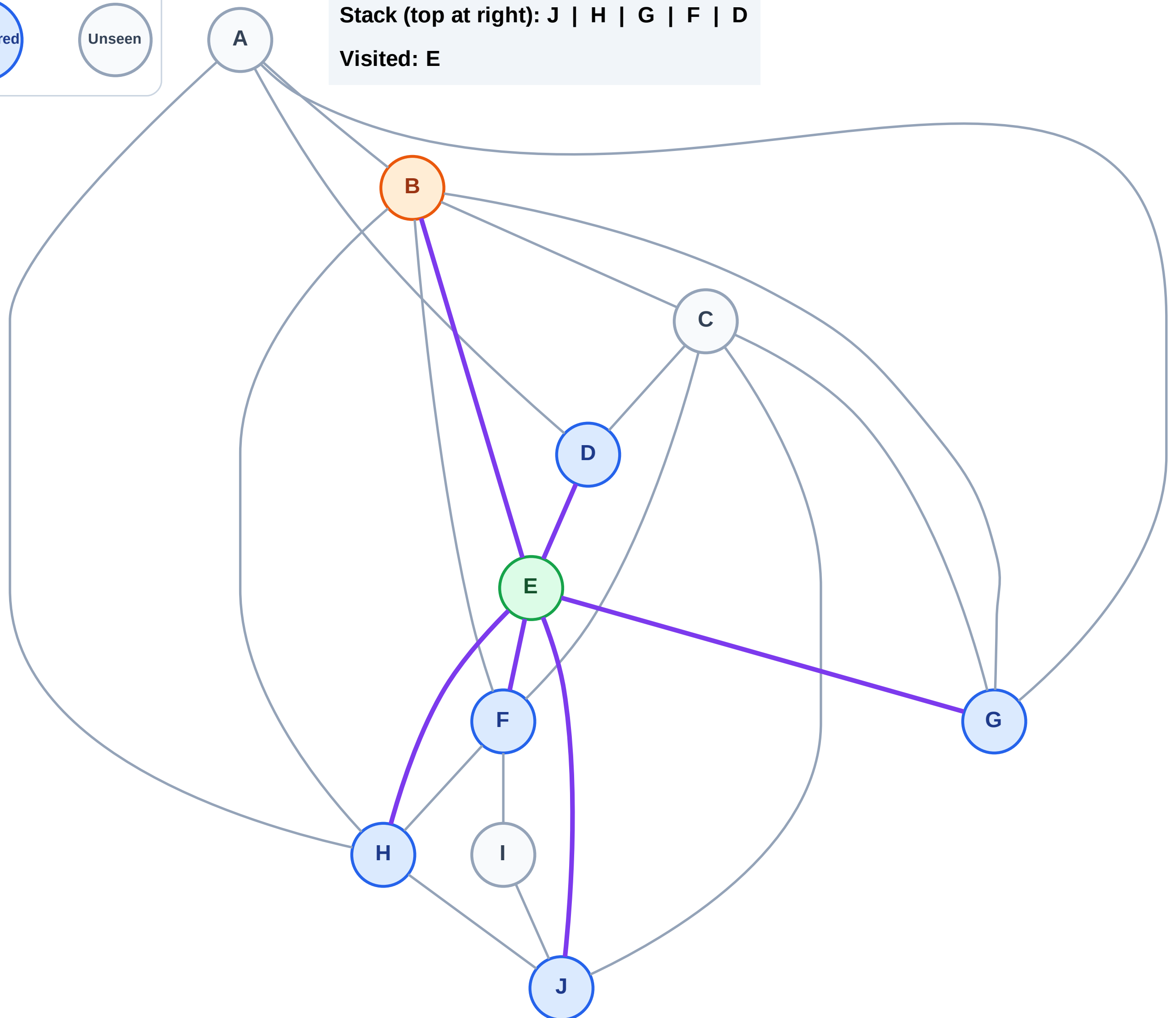
Step 4: Process node B

Legend



Stack (top at right): J | H | G | F | D

Visited: E



Step 5: Discover C, A from B

Step 5: Discover C, A from B



Complete



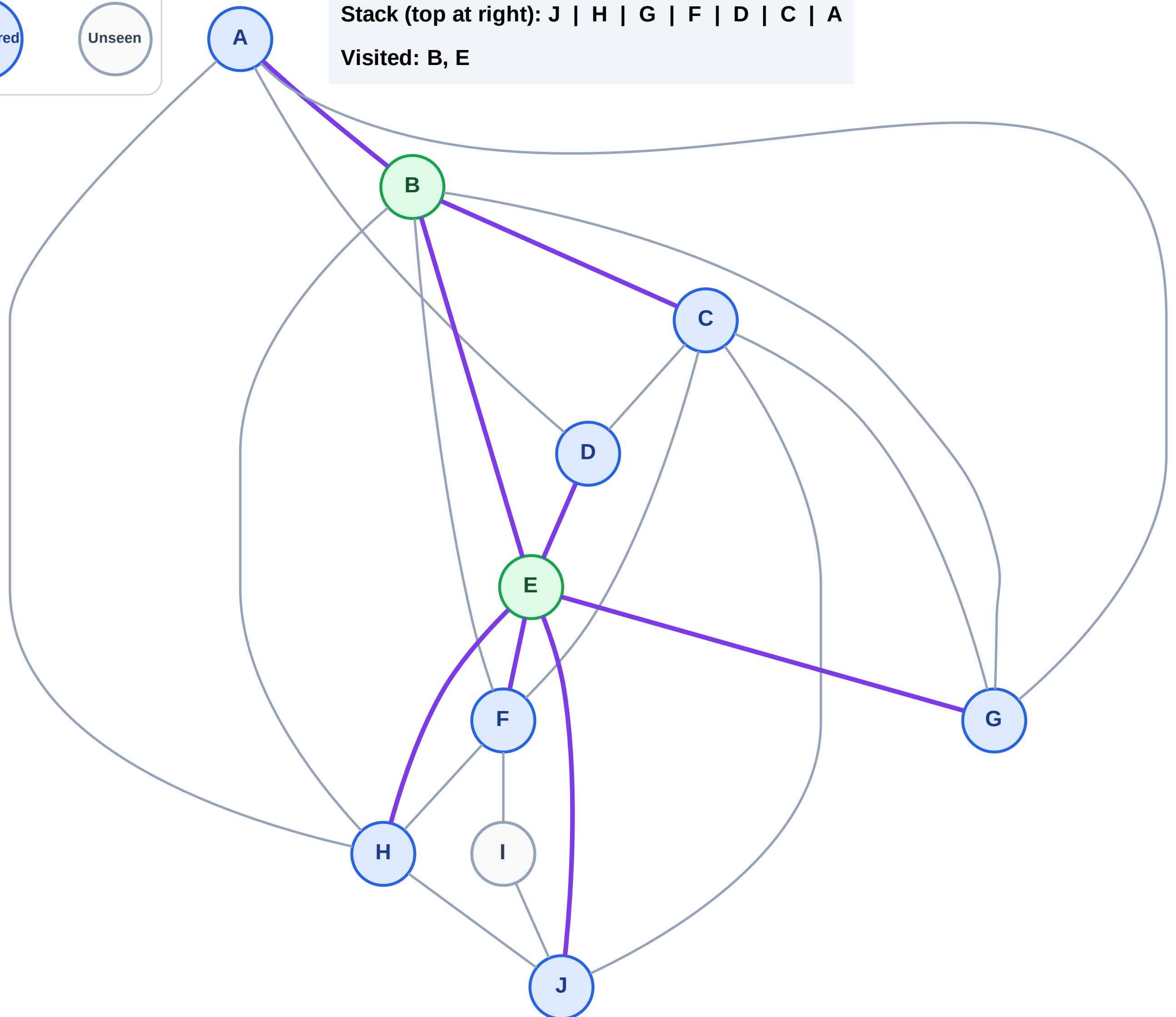
Processing



Discovered

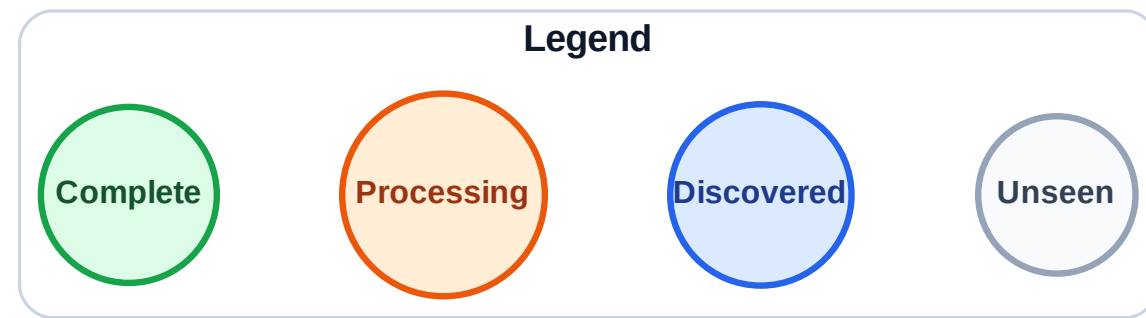


Visited: B, E



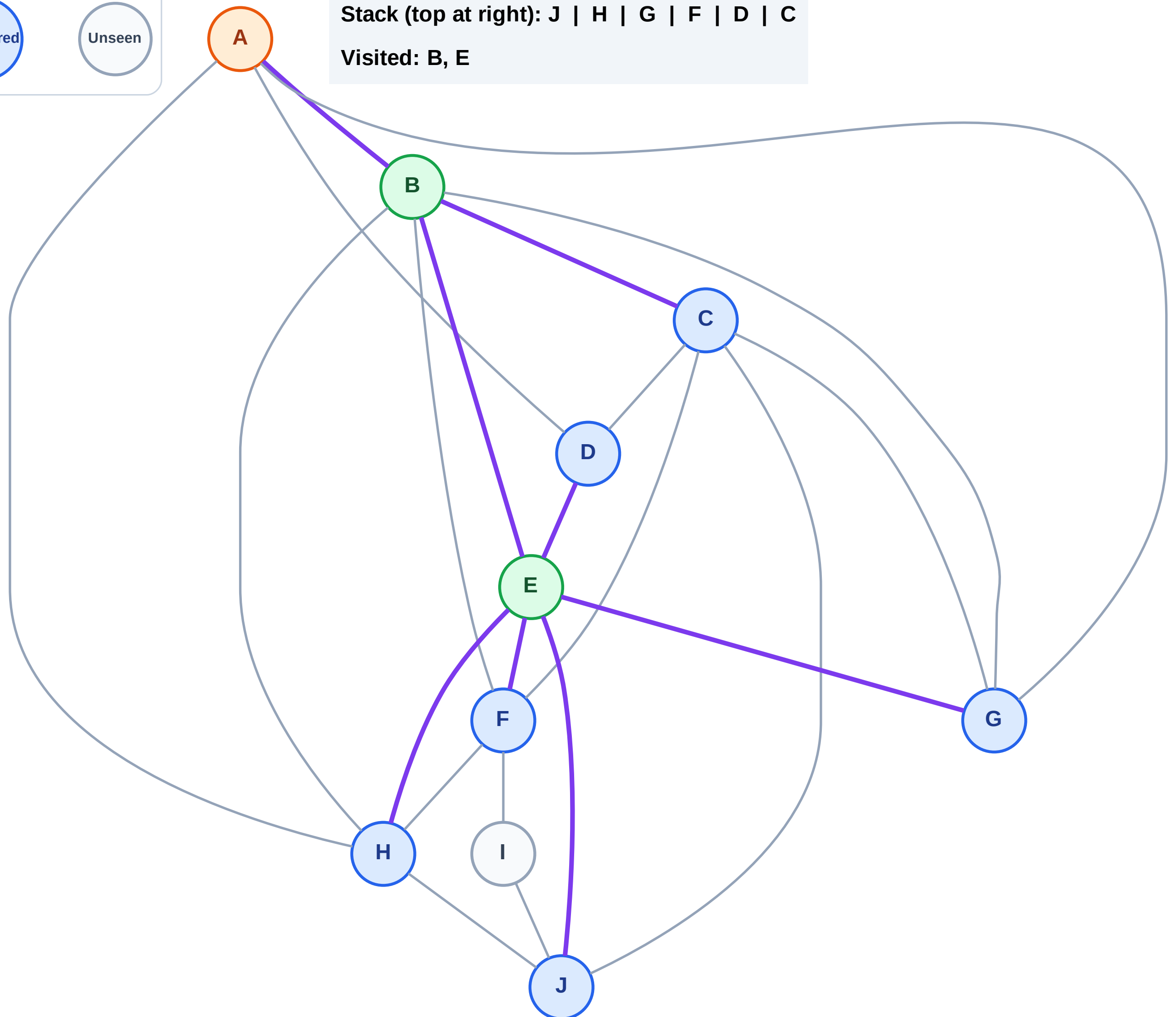
DFS Traversal

Step 6: Process node A



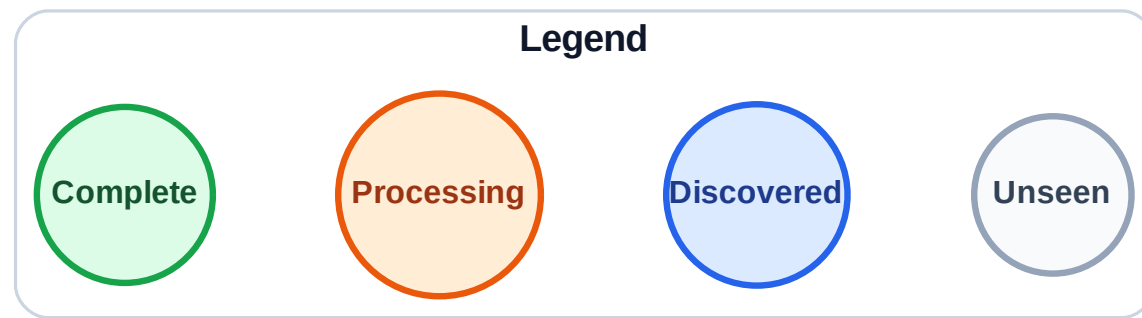
Stack (top at right): J | H | G | F | D | C

Visited: B, E



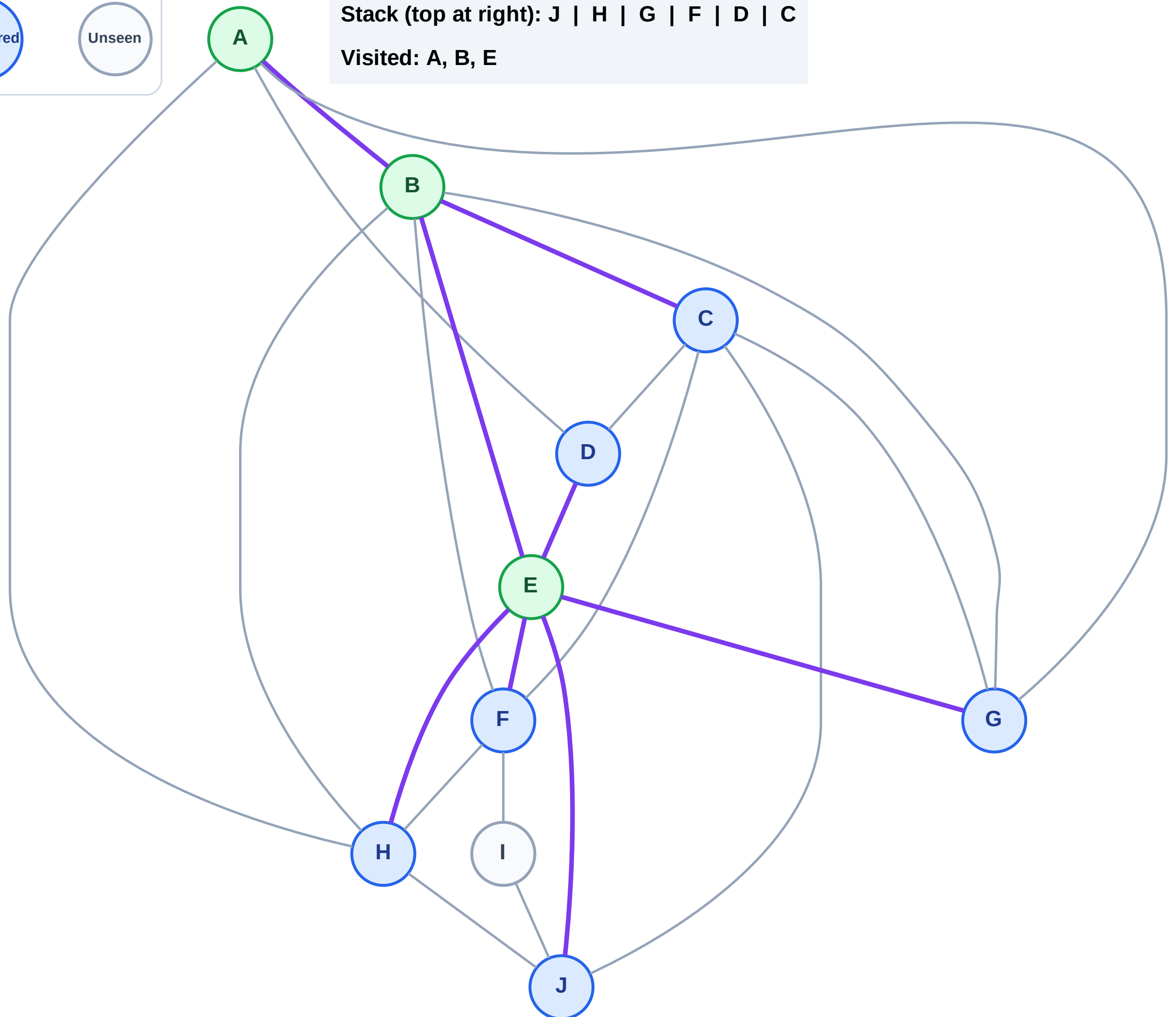
DFS Traversal

Step 7: No new neighbors from A



Stack (top at right): J | H | G | F | D | C

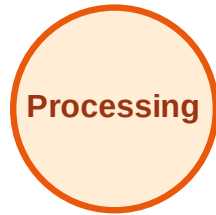
Visited: A, B, E



DFS Traversal

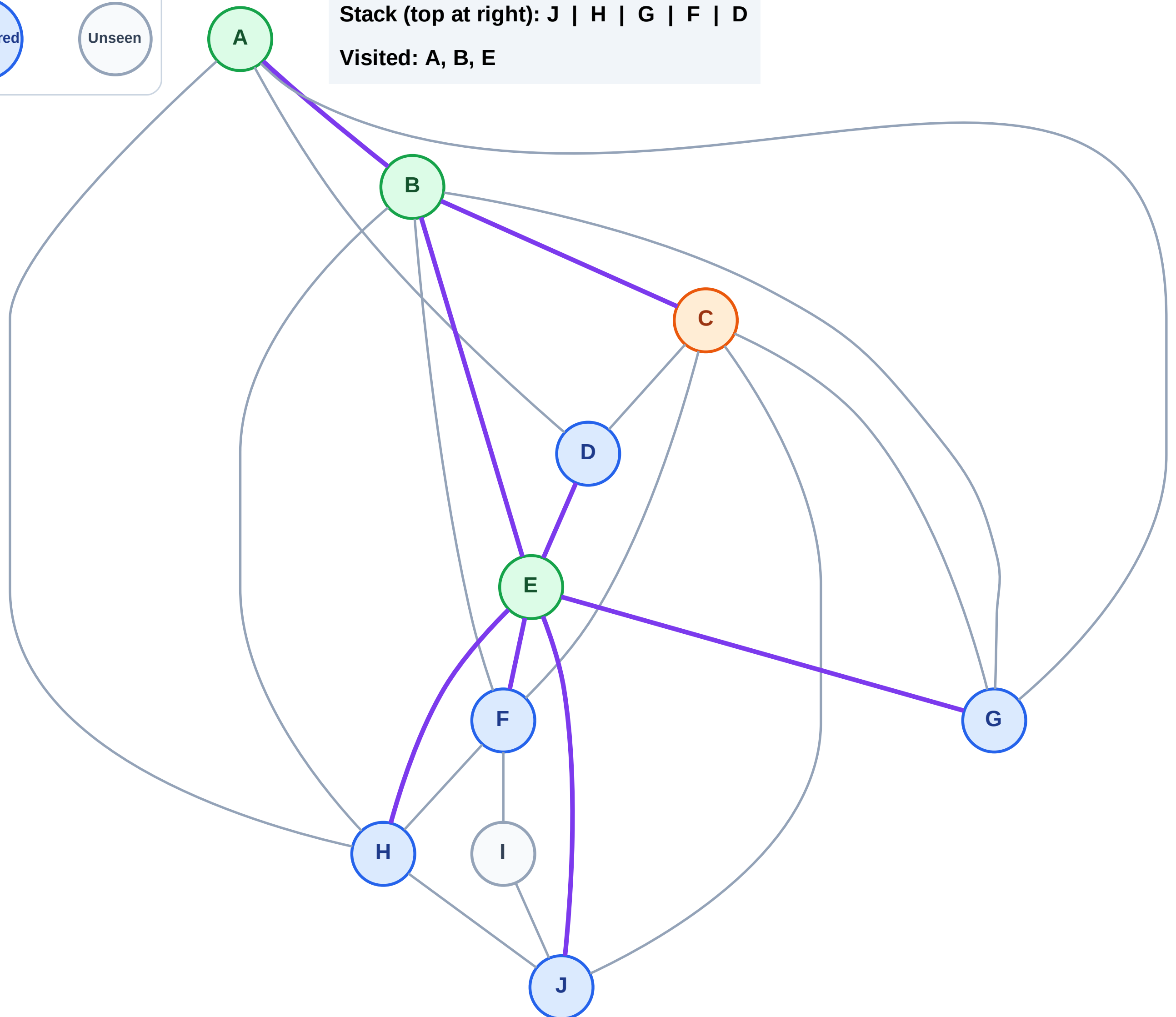
Step 8: Process node C

Legend



Stack (top at right): J | H | G | F | D

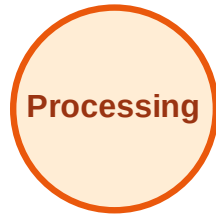
Visited: A, B, E



DFS Traversal

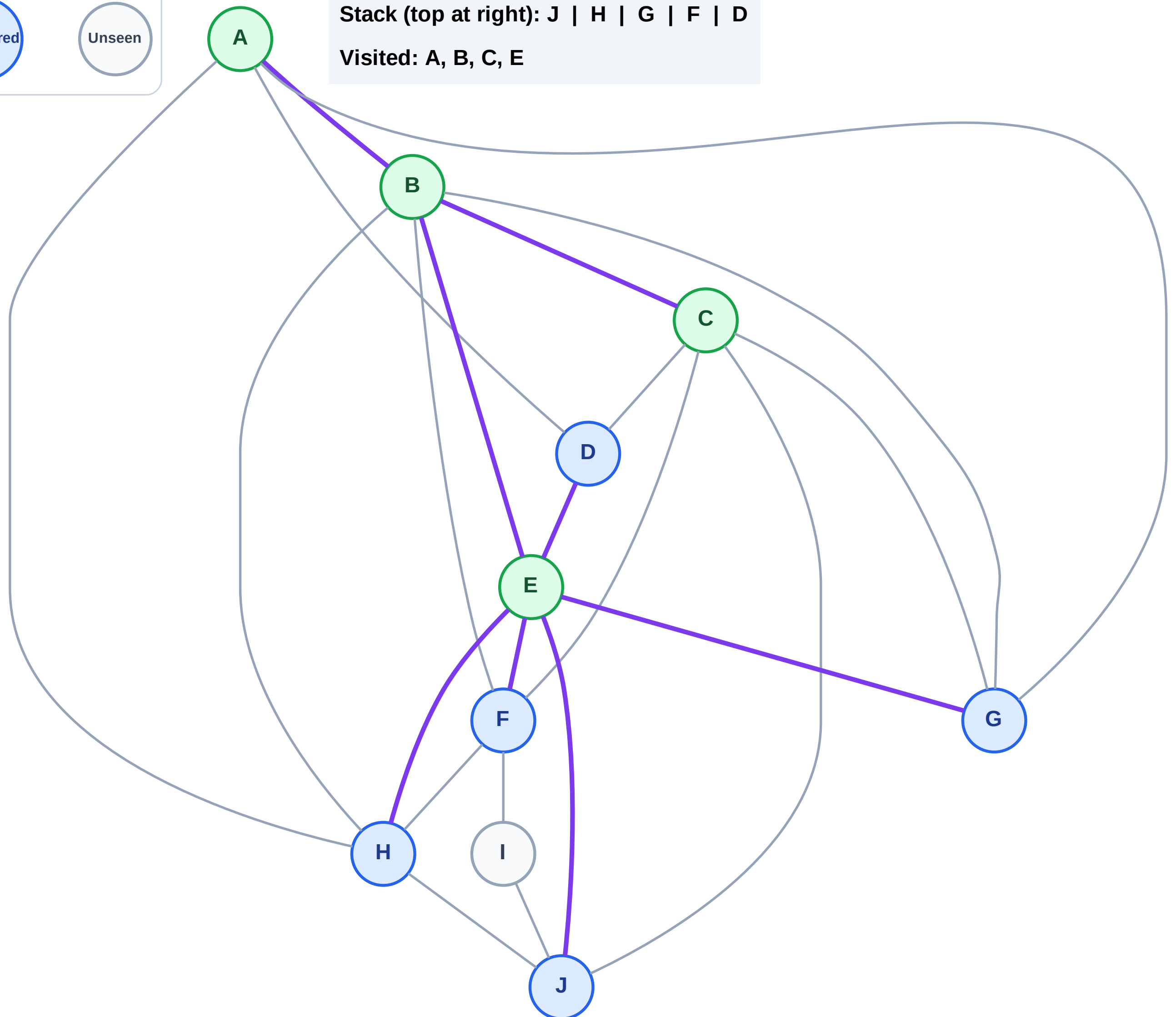
Step 9: No new neighbors from C

Legend



Stack (top at right): J | H | G | F | D

Visited: A, B, C, E



DFS Traversal

Step 10: Process node D

Legend

Complete

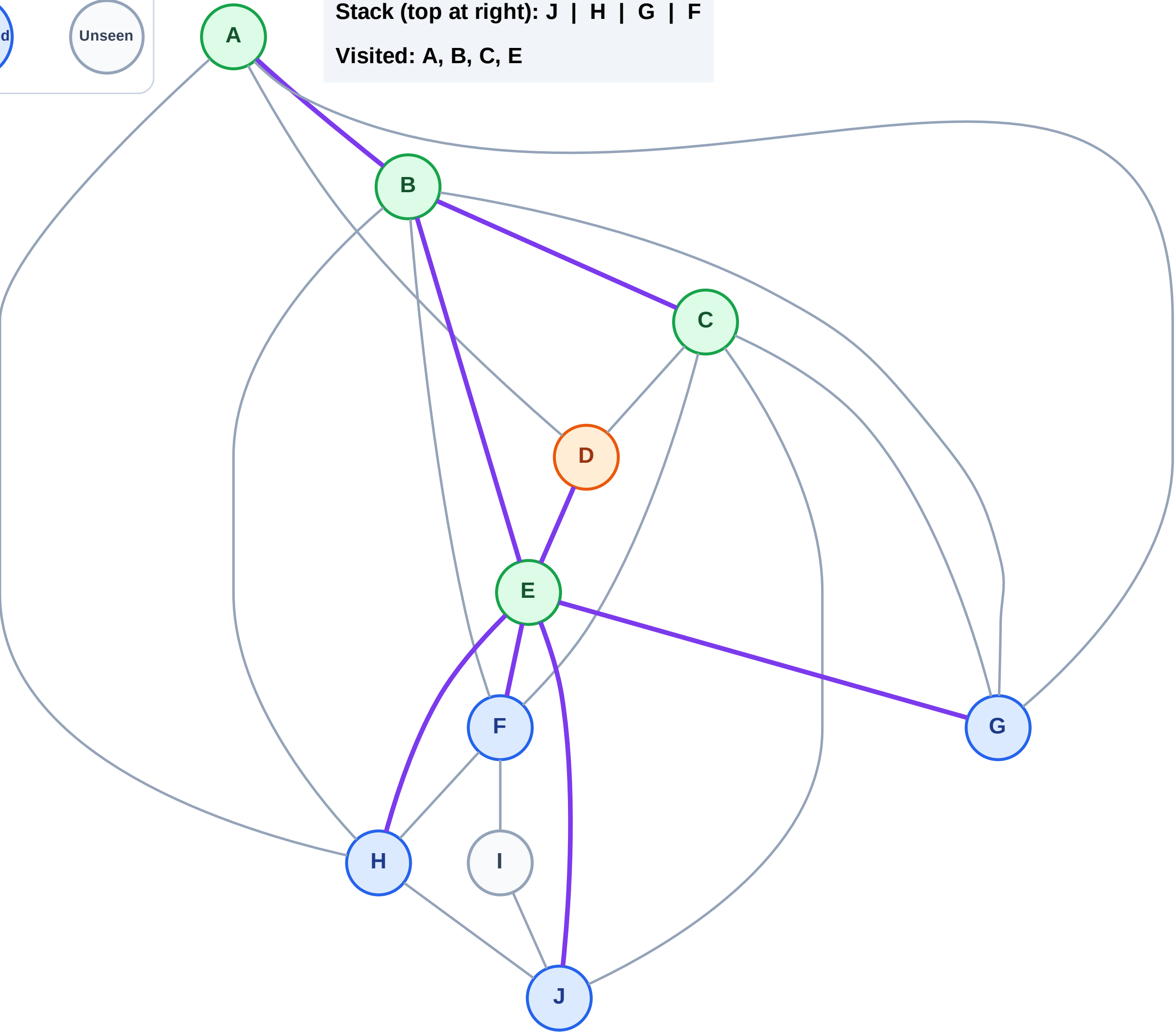
Processing

Discovered

Unseen

Stack (top at right): J | H | G | F

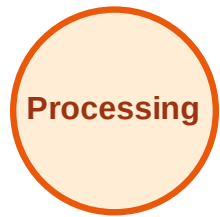
Visited: A, B, C, E



DFS Traversal

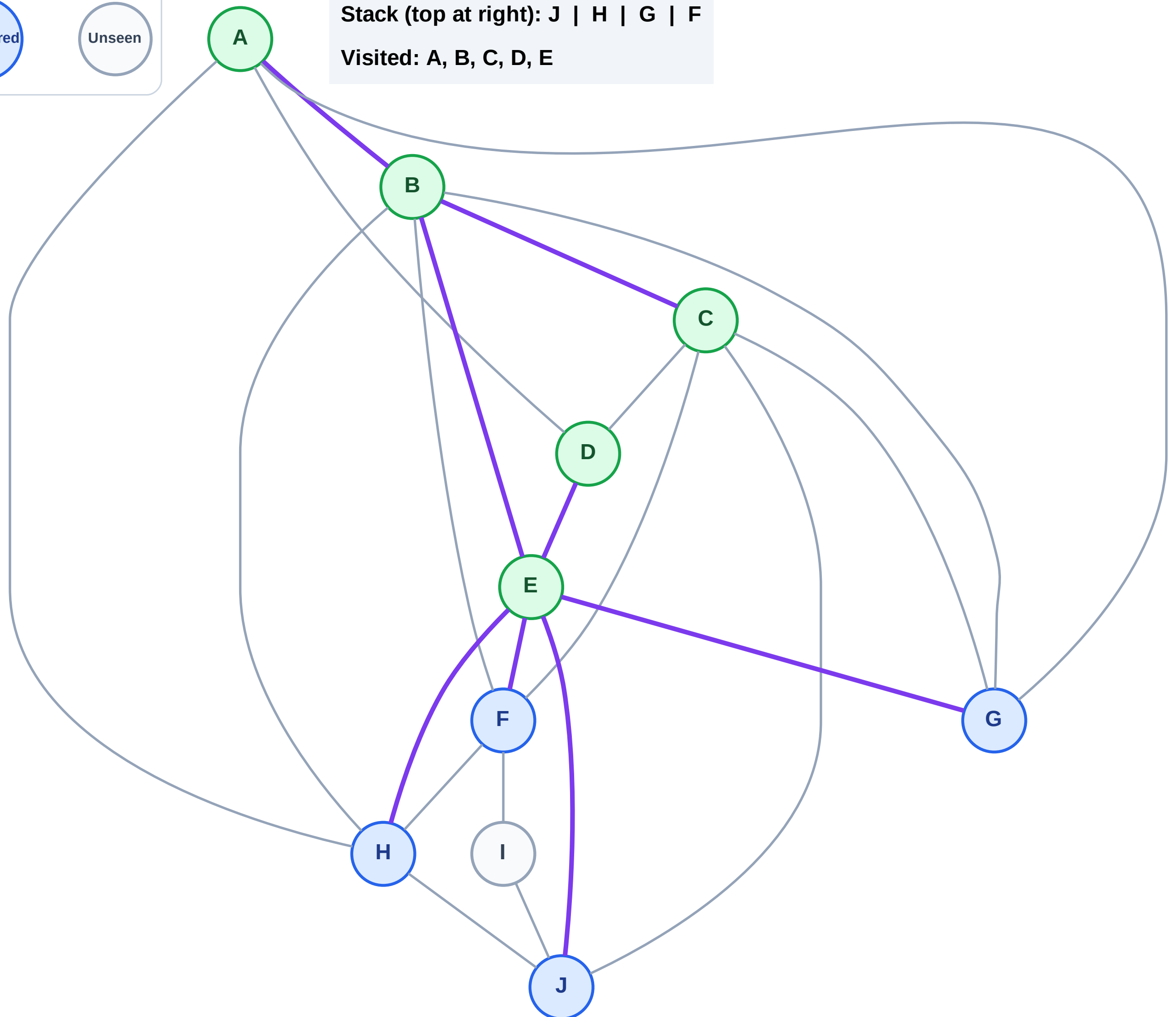
Step 11: No new neighbors from D

Legend



Stack (top at right): J | H | G | F

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node F

Legend

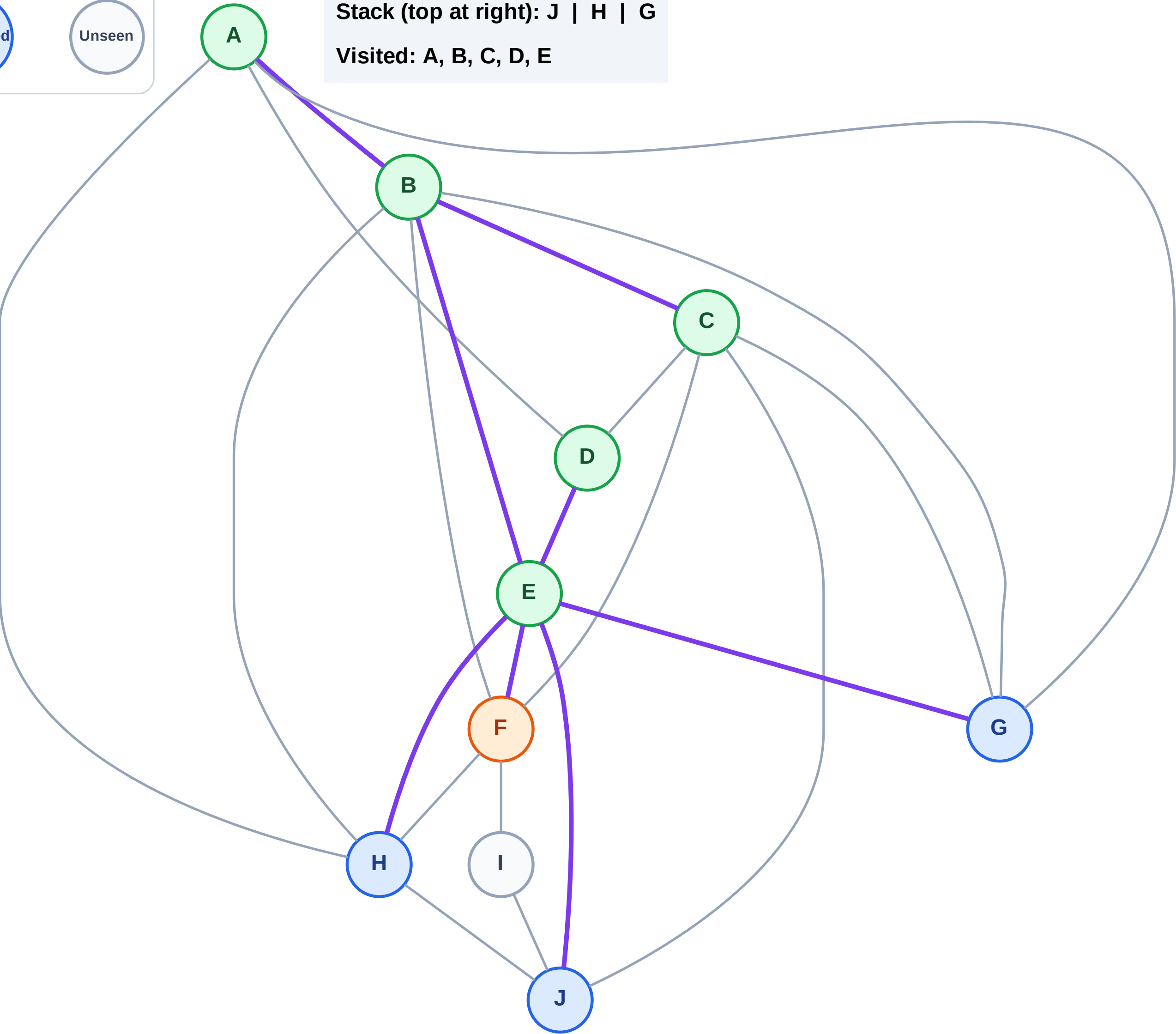
Complete

Processing

Discovered

Unseen

Stack (top at right): J | H | G
Visited: A, B, C, D, E



DFS Traversal

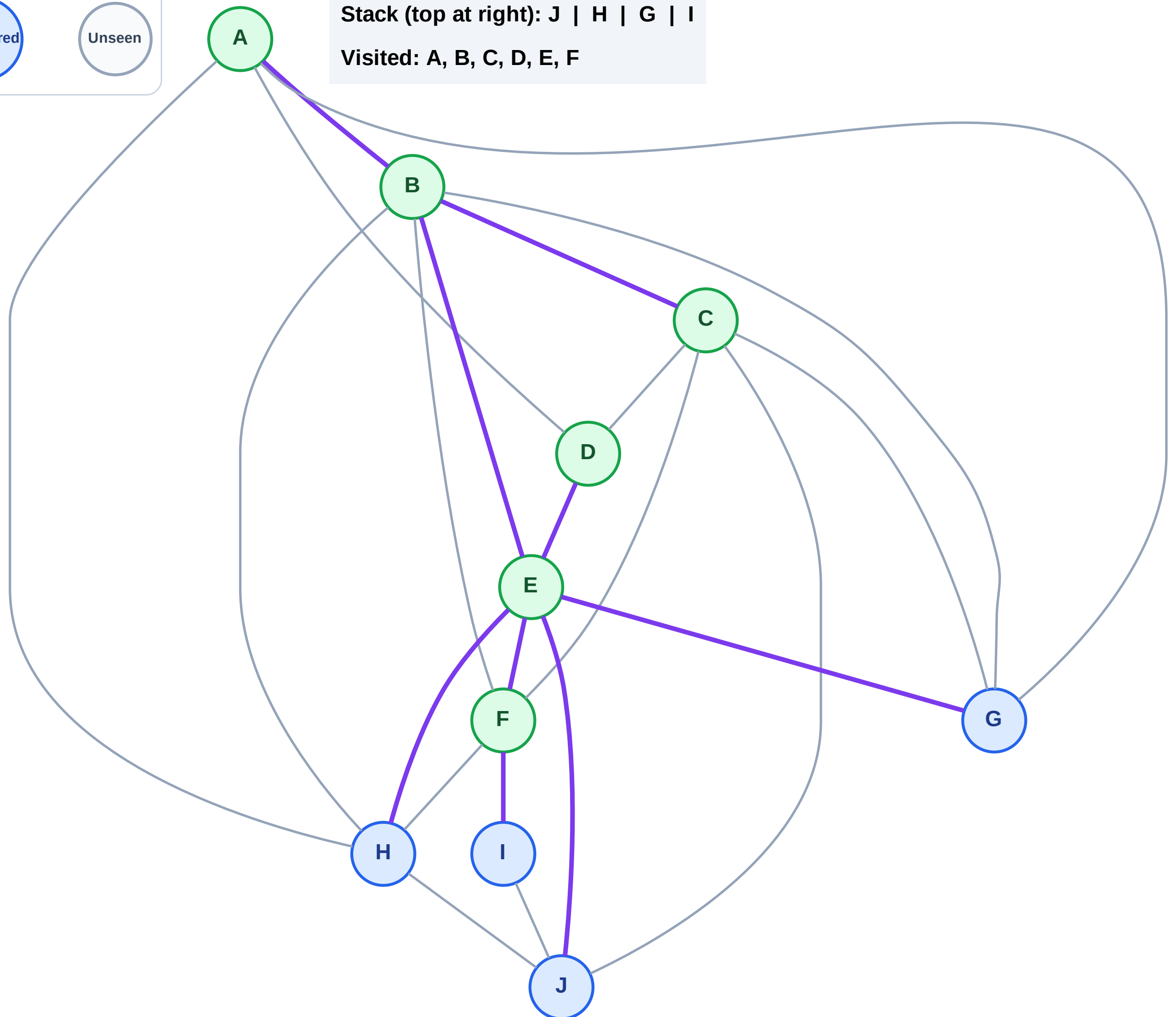
Step 13: Discover I from F

Legend



Stack (top at right): J | H | G | I

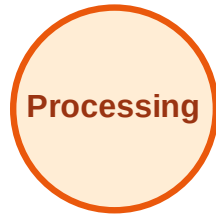
Visited: A, B, C, D, E, F



DFS Traversal

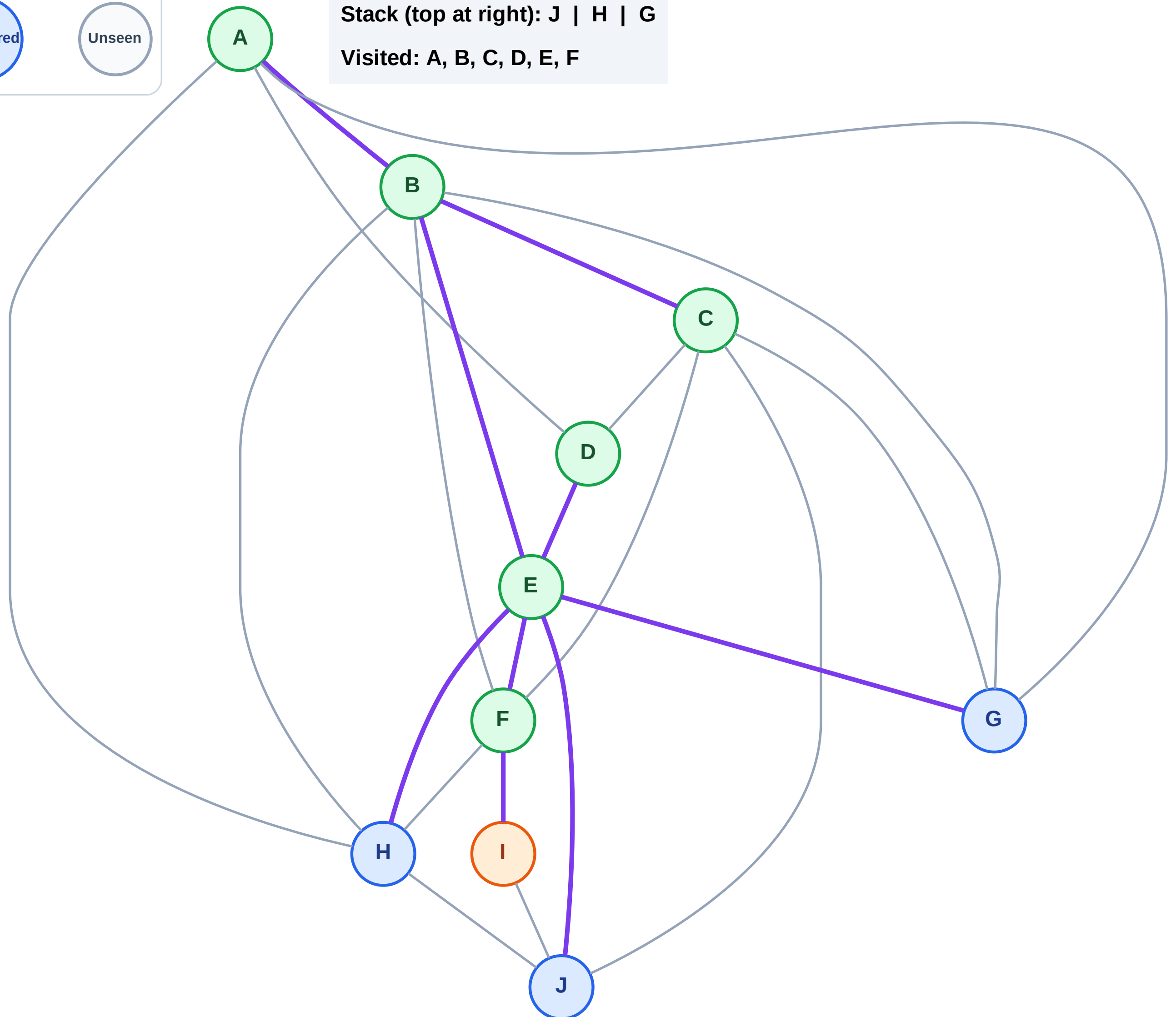
Step 14: Process node I

Legend



Stack (top at right): J | H | G

Visited: A, B, C, D, E, F



DFS Traversal

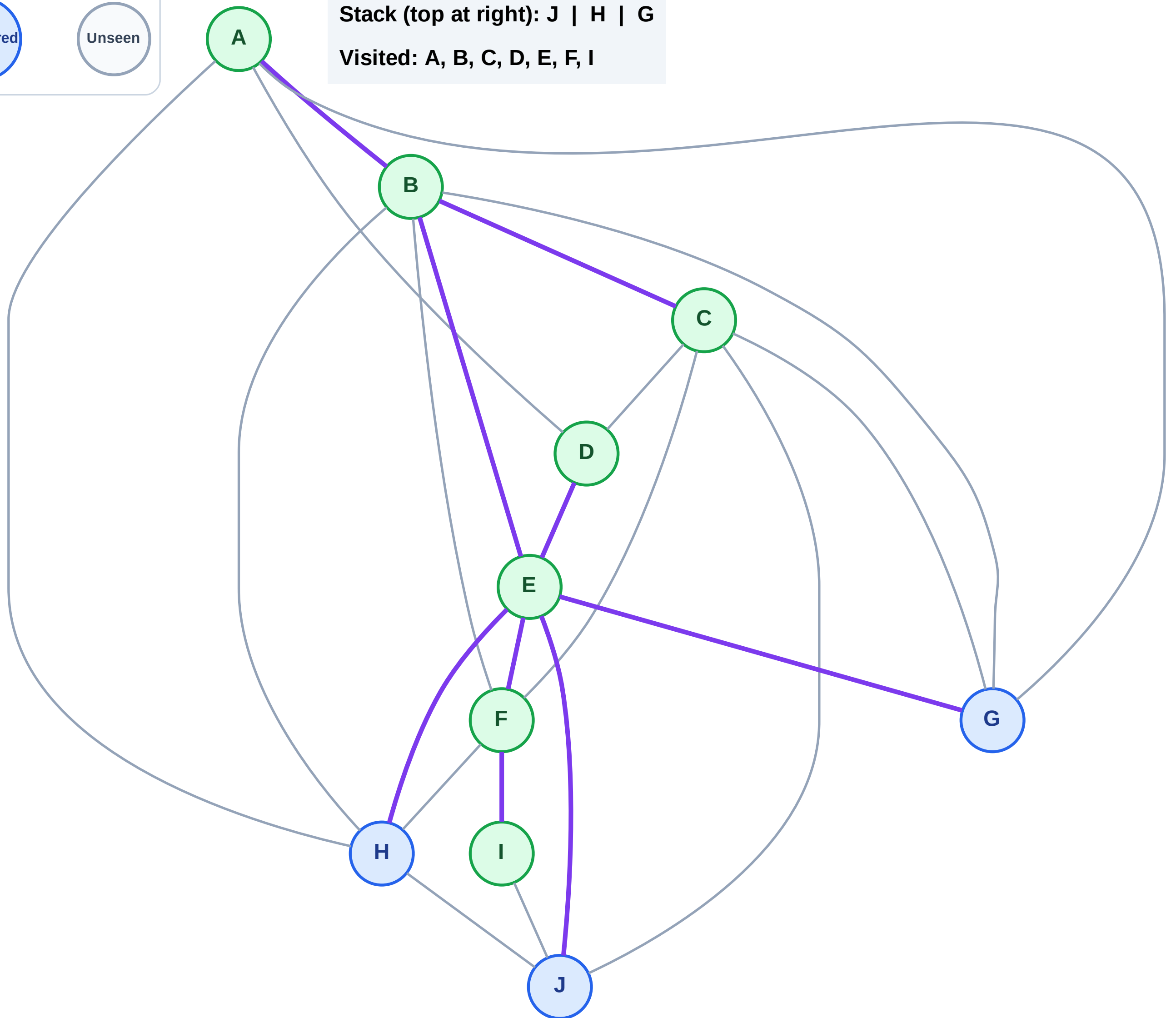
Step 15: No new neighbors from I

Legend



Stack (top at right): J | H | G

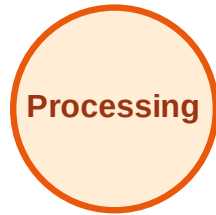
Visited: A, B, C, D, E, F, I



DFS Traversal

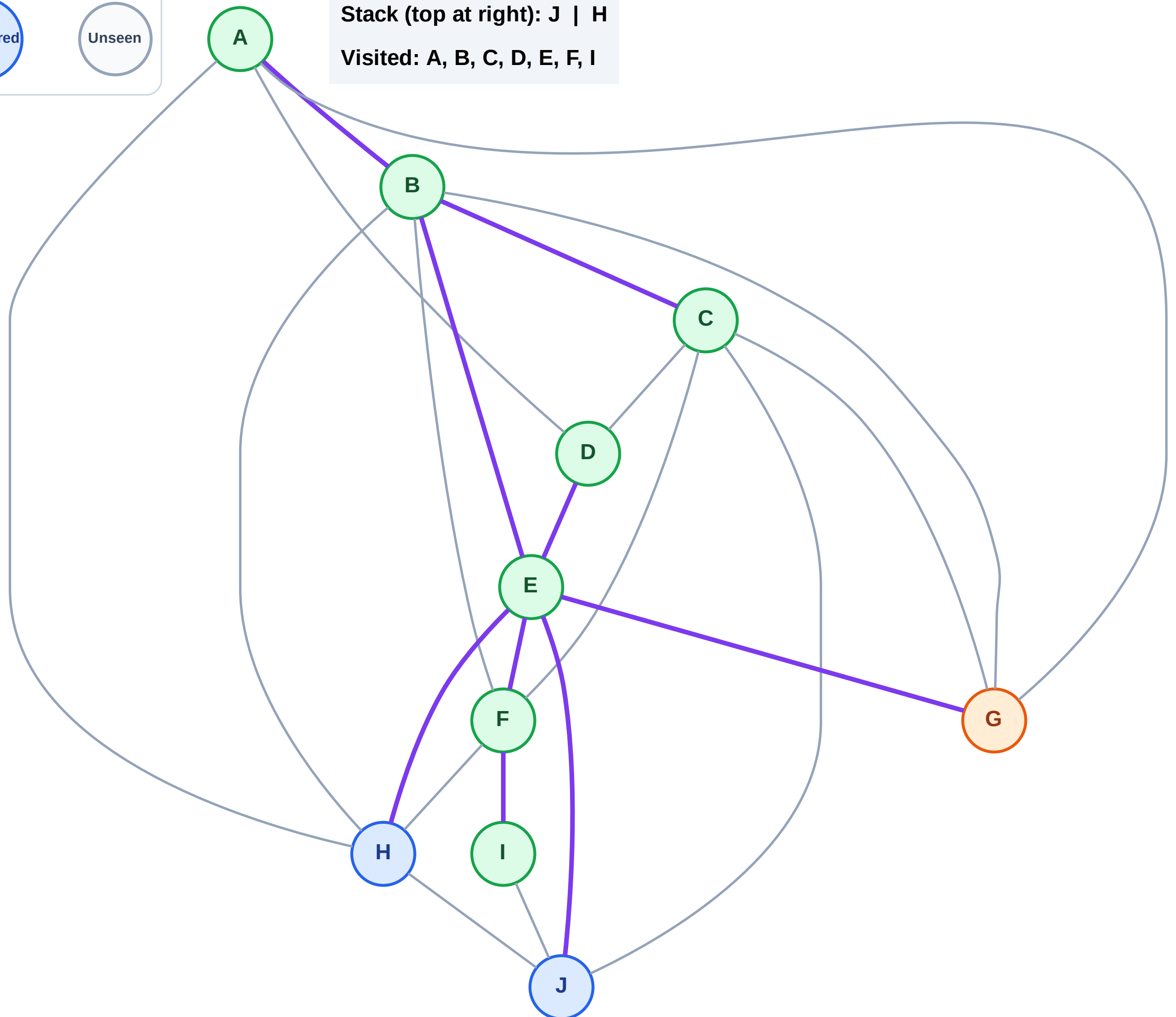
Step 16: Process node G

Legend



Stack (top at right): J | H

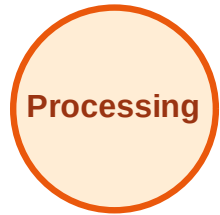
Visited: A, B, C, D, E, F, I



DFS Traversal

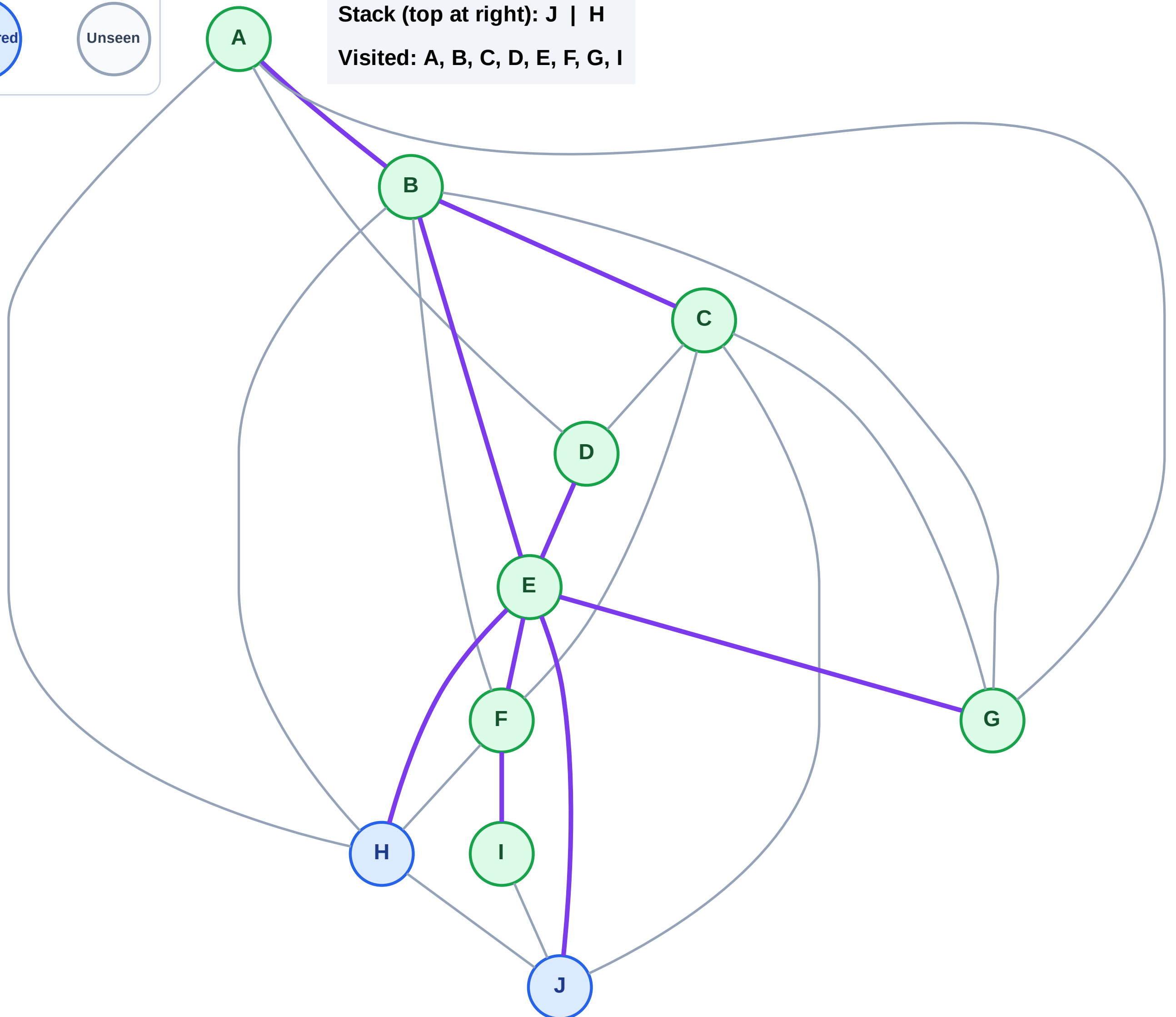
Step 17: No new neighbors from G

Legend



Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



DFS Traversal

Step 18: Process node H

Legend

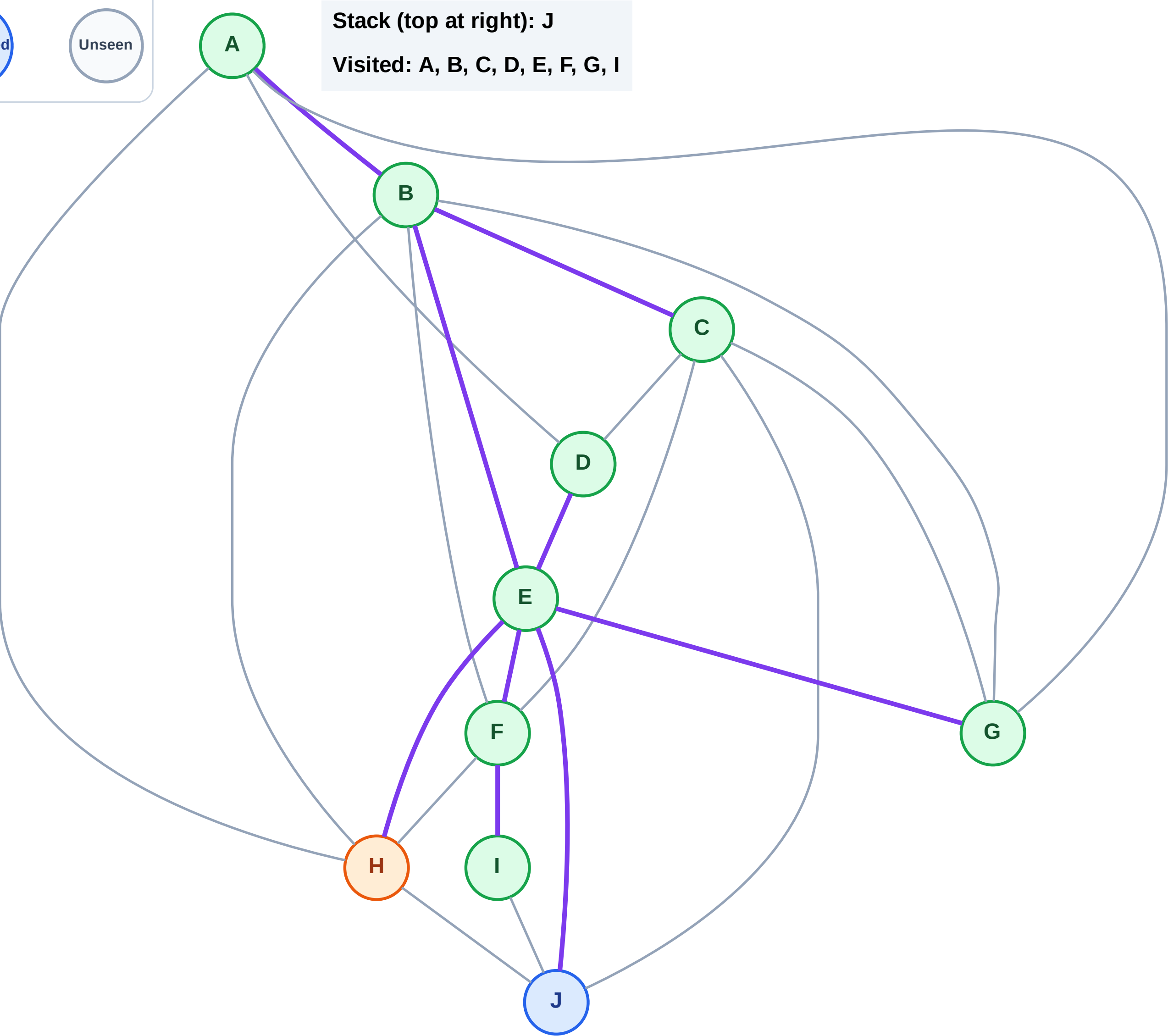
Complete

Processing

Discovered

Unseen

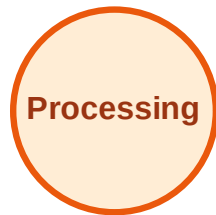
Stack (top at right): J
Visited: A, B, C, D, E, F, G, I



DFS Traversal

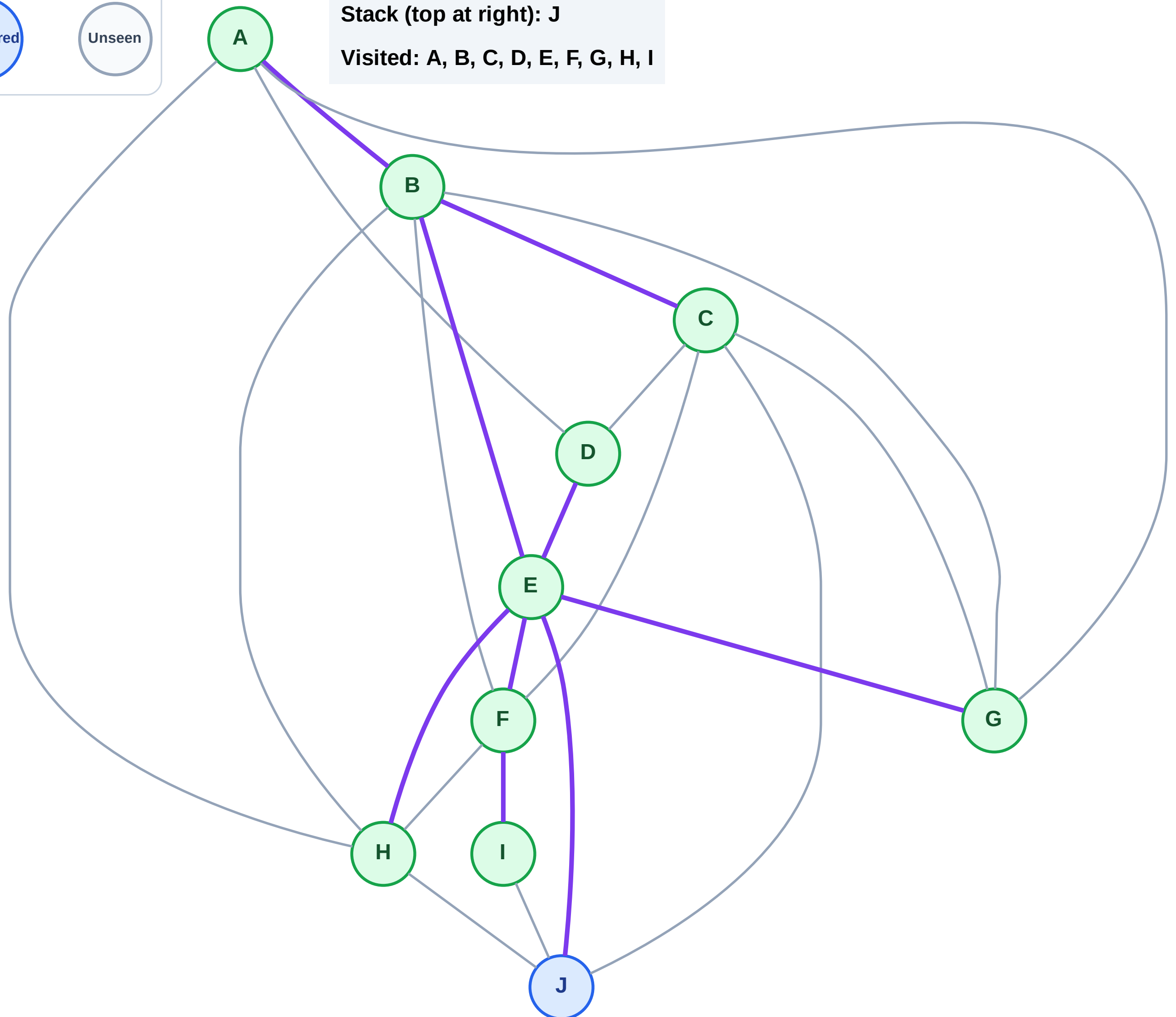
Step 19: No new neighbors from H

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

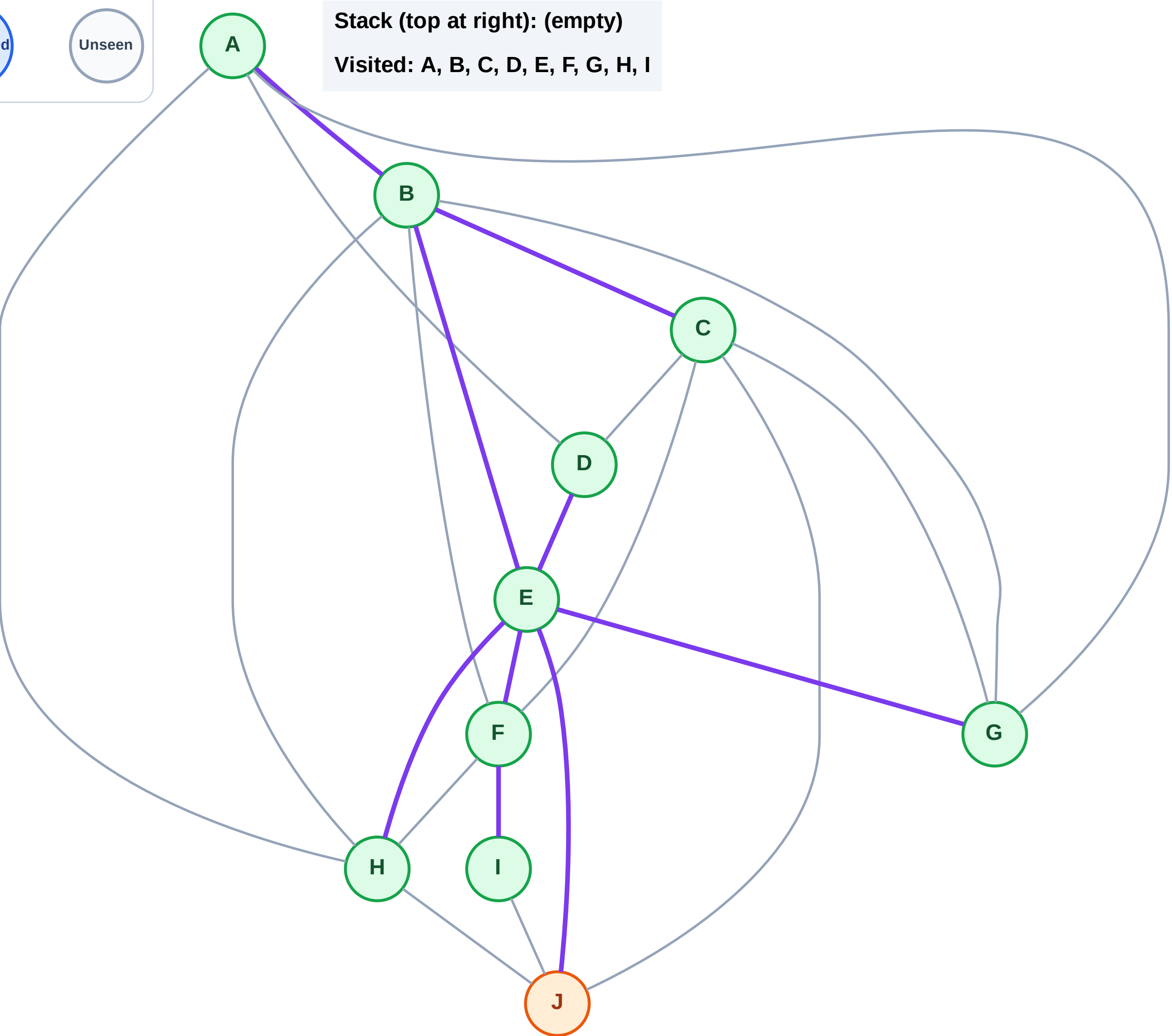
Complete

Processing

Discovered

Unseen

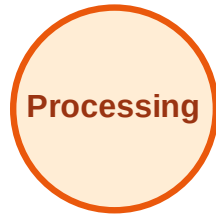
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

